

EDUCATIVE COMMENTARY ON ISI 2021 MATHEMATICS PAPERS

(Last Revised on 14/02/2022)

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Introduction

Commentaries of ISI B.Stat. Entrance Tests began in 2016 because it was the year in which some exciting questions came to the notice of the author. Later on commentaries from 2014 were added. Unfortunately, after 2017, there was a noticeable decline in the quality of questions. Some readers also brought to the notice of the author that a few questions which were hailed in the commentary were picked from some reputed international contests. This did not detract from the intrinsic quality of the questions. Nor is it likely that some candidates had seen the solutions already. But apart from loss of originality, it is an insult to a good problem (for which originally ample time was allowed) to expect the candidates to crack it in a couple of minutes.

Still, the annual commentaries on ISI Entrance Test continue more as a mission, because some students pointed out that unlike the JEE, there are no sources to get the answers of ISI Entrance tests.

Like other annual competitive entrance examinations at the HSC level, the ISI entrance test for 2021 was delayed because of Covid 19. The test was conducted in July 2021. But even after one month, the official question papers have not been released. The text of the questions given here is from an unofficial source and will be revised when the official papers are released, in case there are any discrepancies.

As in the other commentaries, unless otherwise stated, all the references are to the author's *Educative JEE Mathematics*, Third Edition, published by Universities Press, Hyderabad.

I am thankful to the persons who made the question papers available to me. Siddhesh Naik provided the solution to Problem 15 of Paper 1 and also a solution to Problem 1 of Paper 1 based on the second principle of induction which led to the solution given here based on the extremal principle. S. Venkatesh Provided the solution to Problem 19 of Paper 1 and to Problem 2 of Paper 2. Deepanshu

Rajvanshi suggested the alternate derivation of $abc < 4$ given at the end of the solution to Q.7 of Paper 2.

Readers are invited to send their comments and point out errors, if any, in the commentary. They may be sent either by e-mail (kdjoshi314@gmail.com) or by an SMS or a WhatsApp message on mobile (9819961036).

ISI BStat-BMath-UGA-2021 Paper 1 with Comments

N.B. Each question has four options of which ONLY ONE is correct. There are 4 marks for a correctly answered, 0 marks for an incorrectly answered question and 1 mark for each unattempted question.

- Q.1 The number of ways one can express $2^2 3^3 5^5 7^7$ as a product of two numbers a and b where $\gcd(a, b) = 1$, and $1 < a < b$ is

(A) 5 (B) 6 (C) 7 (D) 8

is

Answer and Comments: (C). The numbers 2, 3, 5 and 7 are distinct primes. Since a, b are factors of the given number, say, $n = 2^2 3^3 5^5 7^7$, each has a prime factorisation involving only powers of these four primes. Moreover, as they are relatively prime, no prime can appear in both. So each of the four prime powers is a factor of exactly one of a and b . Since $a < b$ and 7^7 is bigger than the product of all the remaining three prime powers, 7^7 must divide b . As for a , its factors are any combination of these three prime powers. So the problem reduces to finding the number of all subsets of the three element set $\{2^2, 3^3, 5^5\}$. There are in all $2^3 = 8$ such subsets. But we must avoid the empty subset which would give $a = 1$ which is not allowed. So there are 7 possible values of a . Since b is uniquely determined by a (because $ab = 2^2 3^3 5^5 7^7$) 7 is also the number of desired factorisation. ■

A very simple problem based on unique prime power factorisation of positive integers. The paper-setters have been kind to give the prime power factorisation of n . If the candidates were to find it, it would have been a torturous problem.

- Q.2 The sum of all the solutions of $2 + \log_2(x - 2) = \log_{(x-2)} 8$ in the interval $(2, \infty)$ is

(A) $\frac{35}{8}$ (B) 5 (C) $\frac{49}{8}$ (D) $\frac{55}{8}$

Answer and Comments: (C). Let us first rewrite the equation using the substitution

$$u = x - 2 \tag{1}$$

Using the standard identities for the logarithm and keeping in mind that $8 = 2^3$, we get

$$2 + \log_2 u = 3 \log_u 2 = \frac{3}{\log_2 u} \tag{2}$$

If we call $\log_2 u$ as v , this gives a quadratic in v , viz. $2v + v^2 = 3$ or $v^2 + 2v - 3 = 0$, whose roots are $v = 1$ and $v = -3$. Since $u = 2^v$, the corresponding values of u are 2 and $\frac{1}{8}$. Finally, by (1), the respective values of x are 4 and $\frac{17}{8}$. They add to $\frac{49}{8}$. ■

Another absolutely straightforward problem requiring only some elementary identities about logarithms. The quadratic part can be solved just by inspection.

Q.3 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f(x+1) = \frac{1}{2}f(x) \quad (1)$$

for all $x \in \mathbb{R}$ and let $a_n = \int_0^n f(x)dx$ for all integers $n \geq 1$. Then

- (A) $\lim_{n \rightarrow \infty} a_n$ exists and equals $\int_0^1 f(x)dx$
- (B) $\lim_{n \rightarrow \infty} a_n$ does not exist
- (C) $\lim_{n \rightarrow \infty} a_n$ exists if and only if $|\int_0^1 f(x)dx| < 1$
- (D) $\lim_{n \rightarrow \infty} a_n$ exists and equals $2 \int_0^1 f(x)dx$.

Answer and Comments: (D). Had the factor $\frac{1}{2}$ in the R.H.S. of (1) been missing, f would be a periodic function with period 1 and then its integral over any interval of length 1 would be the same. In the present problem, this is not quite so. Still, since the values of f on $[1, 2]$ are half those over the interval $[0, 1]$, we do have

$$\int_1^2 f(x)dx = \frac{1}{2} \int_0^1 f(x)dx \quad (2)$$

(For a formal proof, the substitution $y = x - 1$ transforms the first integral to $\int_0^1 f(y+1)dy = \frac{1}{2} \int_0^1 f(y)dy$ which is exactly $\frac{1}{2} \int_0^1 f(x)dx$ by a replacement of the dummy variable of integration.) Similarly,

$$\int_2^3 f(x)dx = \frac{1}{2} \int_1^2 f(x)dx \quad (3)$$

Continuing, we get a geometric progression of integrals over intervals of the form $[k, k+1]$ for $k = 0, 1, 2, 3, \dots$. By the additivity property of

integrals,

$$\begin{aligned} a_n &= \int_0^n f(x)dx \\ &= \sum_{k=1}^n \int_{k-1}^k f(x)dx \end{aligned} \quad (4)$$

which is a geometric series with common ratio $\frac{1}{2}$ and initial term $\int_0^1 f(x)dx$.

The series converges to $2 \int_0^1 f(x)dx$, which, by definition, is $\lim_{n \rightarrow \infty} a_n$. ■

A simple problem once the idea of writing a_n as a sum of integrals which form a G.P. strikes.

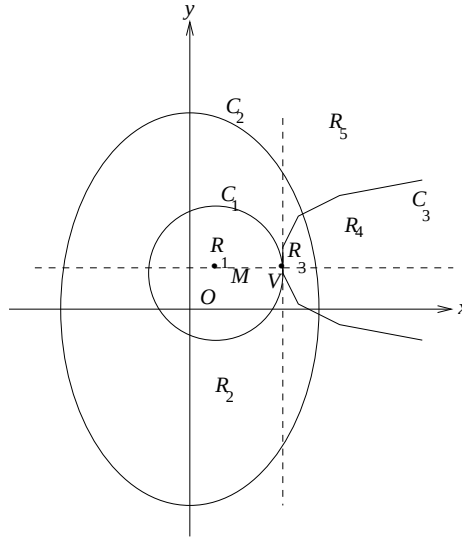
- Q.4 Consider the curves $x^2 + y^2 - 4x - 6y - 12 = 0$, $9x^2 + 4y^2 - 900 = 0$ and $y^2 - 6y - 6x + 51 = 0$. The maximum number of disjoint regions into which these curves divide the XY -plane (excluding the curves themselves) is

(A) 4 (B) 5 (C) 6 (D) 7

Answer and Comments: (B). The statement of the problem is misleading. The three curves are specific curves and so the number of regions they decompose the plane into is some fixed integer. There is nothing like its maximum value. If the three curves were variable members of three classes of curves, then the number of regions would also vary and then there is some point in asking what maximum value it can take. For example, every straight line divides the plane into two regions. Two distinct lines divide it into 3 or 4 parts depending upon whether they are parallel or not. Similarly, a circle and a straight line divide the plane into 3 or 4 parts depending upon whether they intersect in two distinct points or not. Two circles divide the plane into 4 regions when they intersect in two distinct points but only into three regions otherwise.

With three curves the picture gets more complicated. For example, three lines can divide the plane into at most 7 (and not 8) regions, since one of the regions formed by two lines will not be divided by the third line. (See Exercise (1.36) for the case of n straight lines in the plane.)

Coming to the problem, call the three curves as C_1, C_2 and C_3 respectively. By completing squares, C_1 is a circle of radius 5 centred at $(2, 3)$. The second equation can be written as $\frac{x^2}{100} + \frac{y^2}{225} = 1$ and so C_2 is an ellipse with semi-major axis 15 and semi-minor axis 10. (Note that the major axis is vertical.) Rewriting the third equation as $(y - 3)^2 = 6(x - 7)$, we see that C_3 is a parabola with vertex at $(7, 3)$ and the line $y = 3$ as its axis. All the three curves are shown in the figure below.



Note that the centre, say, $M = (2, 3)$ of the circle C_1 lies on the axis of the parabola. Moreover the vertex, say $V = (7, 3)$ of the parabola lies on the circle C_1 . The circle and the parabola have a common vertical tangent at V . There are five regions into which the plane is divided by these three curves : the region R_1 inside C_1 , the region R_2 bounded by the circle C_1 , the ellipse C_2 and the parabola C_3 , the region R_3 inside the ellipse and the parabola, the region R_4 outside the ellipse but inside the parabola and finally, the region R_5 which is outside both the ellipse and the parabola. They are also shown in the diagram. ■

Although conceptually a simple problem, the work involved is considerable. Apart from identifying the three curves, one has to draw them to really see how many regions are formed. The paper setters have given the axes of the ellipse long enough to suggest that the circle will lie entirely within it. But the tangency of the circle and the parabola is not so easy to see. This criticality in the numerical data is crucial for the correct answer. If, for example, the vertex V of the parabola were to lie inside the circle C_1 , then they would intersect in two points and the number of regions would go up by 1. So a qualitative diagram which generally suffices when the curves are in general position will not work here.

Q.5 A box has 13 distinct pairs of socks. Let p_r denote the probability of having at least one matching pair among a bunch of r socks drawn at random from the box. If r_0 is the maximum possible value of r such that $p_r < 1$, then the value of p_{r_0} is

- (A) $1 - \frac{12}{\binom{26}{12}}$ (B) $1 - \frac{13}{\binom{26}{13}}$
 (C) $1 - \frac{2^{13}}{\binom{26}{13}}$ (D) $1 - \frac{2^{12}}{\binom{26}{12}}$

Answer and Comments: (C). By the pigeon hole principle, if more than 13 socks are selected, then two of them are bound to be matching each other. On the other hand, by selecting exactly one from each pair we do get a selection with no matching pair. Hence $r_0 = 13$. To find $p_{r_0} = p_{13}$, it is easier to find the complementary probability, i.e. the probability that there are no matching pairs in the 13 socks picked at random. This can be done in 2^{13} ways, by picking exactly one from each pair. The total number of ways to pick 13 socks from a collection of 26 socks is $\binom{26}{13}$.

Hence the complementary probability, i.e. $1 - p_{13}$ equals $\frac{2^{13}}{\binom{26}{13}}$. Therefore (C) is the correct value of p_{r_0} . ■

A simple problem. The very format of all options suggests that complementary probability is involved.

Q.6 Let $a, b, c, d > 0$ be any real numbers. Then the maximum possible value of $cx + dy$ over all points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

- (A) $\sqrt{a^2c^2 + b^2d^2}$
 (B) $\sqrt{a^2b^2 + c^2d^2}$
 (C) $\sqrt{\frac{a^2c^2 + b^2d^2}{a^2 + b^2}}$
 (D) $\sqrt{\frac{a^2b^2 + c^2d^2}{c^2 + d^2}}$

Answer and Comments: (A). We want to maximise a function $cx + dy$ of two variables x and y . But these two variables are constrained by the requirement

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (1)$$

As a result, this problem comes under what is called **constrained optimisation**. There are several ways to tackle them. The easiest is to parametrise the curve (here the ellipse), in terms of some parameter and thereby express the function to be maximised as a function of one variable, viz. that parameter. Here we take the standard parametrisation of the ellipse

$$x = a \cos \theta, \quad y = b \sin \theta \quad (2)$$

where θ varies over $[0, 2\pi]$. With these substitutions the problem reduces to maximising the function $f(\theta)$ defined by

$$f(\theta) = ac \cos \theta + bd \sin \theta \quad (3)$$

over the interval $[0, 2\pi]$. The calculus method is available for this. But we can also do it using elementary trigonometry. Let α be the (unique) angle in $[0, 2\pi)$ for which

$$\sin \alpha = \frac{ac}{\sqrt{a^2c^2 + b^2d^2}} \quad (4)$$

$$\text{and } \cos \alpha = \frac{bd}{\sqrt{a^2c^2 + b^2d^2}} \quad (5)$$

(Actually, α will lie in $(0, \frac{\pi}{2})$ since a, b, c, d are all positive.) Then

$$\begin{aligned} f(\theta) &= \sqrt{a^2c^2 + b^2d^2} (\sin \alpha \cos \theta + \cos \alpha \sin \theta) \\ &= \sqrt{a^2c^2 + b^2d^2} \sin(\alpha + \theta) \end{aligned} \quad (6)$$

As the maximum possible value of the sine function is 1, the maximum possible value of $f(\theta)$ is $\sqrt{a^2c^2 + b^2d^2}$. ■

A simple problem on constrained optimisation. See Comment No. 14, 15 and 16 of Chapter 13 for more problems and a discussion of other methods.

Q.7 Let $f(x) = \sin x + \alpha x$, $x \in \mathbb{R}$ where α is a fixed real number. The function is one-to-one if and only if

- (A) $\alpha > 1$ or $\alpha < -1$
- (B) $\alpha \geq 1$ or $\alpha \leq -1$
- (C) $\alpha \geq 1$ or $\alpha < -1$
- (D) $\alpha > 1$ or $\alpha \leq -1$

Answer and Comments: (B). If we go strictly by definition, to say that f is one-to-one means that the equation

$$f(x) = f(y) \quad (1)$$

cannot hold when $x \neq y$. For the given function, this translates into

$$\sin x - \sin y = \alpha(y - x) \quad (2)$$

and, with a trigonometric identity, further to

$$2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) = \alpha(y - x) \quad (3)$$

Call $x - y$ as $2h$. Then $h \neq 0$ and dividing by it gives

$$\cos \frac{x+y}{2} \frac{\sin h}{h} = -\alpha \quad (4)$$

Without loss of generality we may suppose that $h > 0$. Then by the well-known inequality

$$\sin \theta < \theta \quad (5)$$

for all $\theta > 0$, the second factor of the L.H.S. of (4) lies in $(0, 1)$. The first factor lies in the interval $[-1, 1]$. Hence the L.H.S. lies in the interval $(-1, 1)$. So for (2) to hold, $\alpha \in (-1, 1)$. Therefore for $\alpha \leq -1$ or $\alpha \geq 1$, (2) cannot hold, i.e. the function f is one-to-one. Hence if at all one of the given options is correct, it is (B).

The same conclusion can be arrived at using derivatives. First, note that a continuous, one to one function has to be either strictly monotonically increasing or strictly monotonically decreasing. (A proof of this result can be given using the IVP (Intermediate Value Property), see Comment No. 6 of Chapter 16.) Conversely strict monotonicity trivially implies one to one. So the problem is reduced to checking for which values of α , the function $f(x) = \sin x + \alpha x$ is strictly increasing/decreasing. It is a popular misconception that for a differentiable function, strictly increasing is equivalent to the positivity of the derivative. But this is not quite so. Surely, by Lagrange mean value theorem, if f' is positive everywhere then f is strictly increasing. But the converse is false. If $f'(x)$ is positive for all $x \in (a, b)$, that is sufficient to conclude that $f(a) < f(b)$ because the point c given by the LMVT lies in (a, b) . So, it may happen that f' vanishes at the end points and still f is strictly increasing on an interval. In fact, we can go further and prove that if the derivative of a function g is non-negative at all points of $[a, b]$ but vanishes at a finite number of points in $[a, b]$, then $g(a) < g(b)$. (See Exercise (13.8). The essential idea is to split the interval $[a, b]$ at the points where g' vanishes and apply LMVT to each subinterval.) This is precisely what happens for the function $f(x) = \sin x + x$. (See again Exercise (13.8).) Hence $\sin x + \alpha x$ is strictly increasing not only for all $\alpha > 1$, but for $\alpha = 1$ too. By a similar reasoning, it is strictly decreasing for all $\alpha \leq -1$. So again (B) is correct among the four options.

However, the question asks for a condition which is necessary as well as sufficient for $\sin x + \alpha x$ to be one-to-one. We have only proved sufficiency, viz. either $\alpha \geq 1$ or $\alpha \leq -1$. So, for the sake of completeness, we must verify that for $\alpha \in (-1, 1)$, f is not one-to-one, i.e. (2) has a solution in which $x \neq y$. This is easy if $\alpha = 0$ because then the function is periodic. However, for other values of $\alpha \in (-1, 1)$, we can only prove the existence of two points $x \neq y$ which satisfy (2). To do this we use derivatives. Let $f(x) = \sin x + \alpha x$. Then $f'(x) = \cos x + \alpha$. Let $x_0 = \cos^{-1}(-\alpha)$. Then $f'(x_0) = 0$, but $f''(x_0) = -\sin x_0 = \pm\sqrt{1 - \alpha^2}$. Since $|\alpha| < 1$, $f''(x_0) \neq 0$.

Depending upon whether it is positive or negative, the function $f(x)$ has a local minimum or a local maximum at x_0 . In either case, there are points x, y on the two sides of x_0 at which $f(x) = f(y)$. So f is not one-to-one.

We have now completely proved that (B) is the correct answer. ■

Those who confuse strictly increasing nature of a function with the positivity of its derivative will give (A) as the correct answer. The problem is a good punishment for them. Those who are familiar with the strict monotonicity of $\sin x + x$ are likely to choose (B). But it is doubtful how many of them realise that for $|\alpha| < 1$, $\sin x + \alpha x$ is not one to one. So this would be an excellent question where reasoning is asked, but not as an MCQ.

Q.8 The value of

$$1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+2021}$$

is

$$(A) \frac{2021}{1010} \quad (B) \frac{2021}{1011} \quad (C) \frac{2021}{1012} \quad (D) \frac{2021}{1013}$$

Answer and Comments: (B). The n -th term of the given series is $\frac{1}{1+2+\dots+n}$ which equals $\frac{2}{n(n+1)}$. Hence the given expression, say E , can be evaluated as

$$E = 2 \sum_{n=1}^{2021} \frac{1}{n(n+1)} \quad (1)$$

The R.H.S. is a standard example of a telescopic series. We write $\frac{1}{n(n+1)}$ as $\frac{1}{n} - \frac{1}{n+1}$. Then

$$E = 2 \sum_{n=1}^{2021} \left[\frac{1}{n} - \frac{1}{n+1} \right] \quad (2)$$

All terms get canceled except $\frac{1}{1}$ in the first and $\frac{1}{2022}$ in the last. So $E = 2(1 - \frac{1}{2022}) = \frac{2021}{1011}$. ■

The formula for $1+2+3+\dots+n$ is too well known. The problem unfolds itself once it is recognised that the series in (1) is a telescopic series. It often takes some skill to recast a series as a telescopic series. But in the present problem, the series involved is usually the very first example of a telescopic series the students learn.

Q.9 The volume of the region $S = \{(x, y, z) : |x| + 2|y| + 3|z| \leq 6\}$ is

- (A) 36 (B) 48 (C) 72 (D) 6

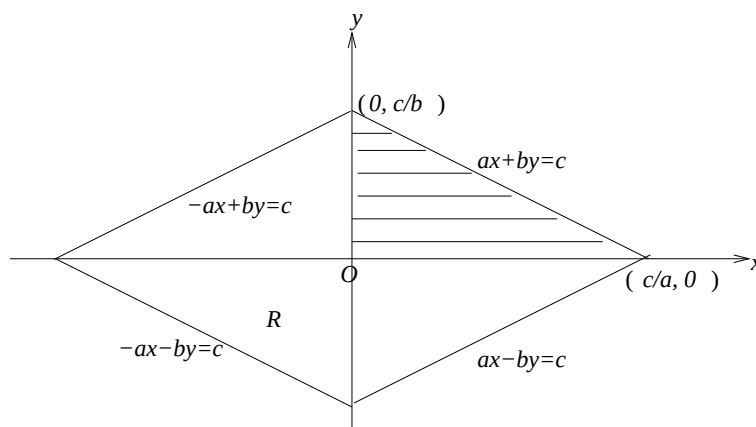
Answer and Comments: (B). We first have to identify the solid (a better word than region) S . It often pays to draw analogies from similar problems in the plane. Thus consider the plane region

$$R = \{(x, y) : a|x| + b|y| \leq c\} \quad (1)$$

where a, b, c are some positive real numbers. This is a parallelogram (actually a rhombus) bounded by the four straight lines

$$\pm ax \pm by = c \quad (2)$$

The region R is shown in the figure on the next page.



The part of the region in the positive quadrant (shown by shading) is a triangle bounded by the two axes and the line

$$ax + by = c \quad (3)$$

Clearly this is a triangle with vertices $(0, 0)$, $(\frac{c}{a}, 0)$ and $(0, \frac{c}{b})$. It is a right angled triangle with area $\frac{1}{2} \frac{c^2}{ab}$. The region R is the union of four congruent copies of this triangle which do not overlap except for the sides. So the area of R is $\frac{2c^2}{ab}$.

In an entirely analogous manner the solid $\{(x, y, z) : a|x| + b|y| + c|z| \leq d\}$ where a, b, c, d are positive real numbers, is the union of eight congruent tetrahedra. The one lying in the positive orthant has its vertices at $(0, 0, 0)$, $(\frac{d}{a}, 0, 0)$, $(0, \frac{d}{b}, 0)$ and $(0, 0, \frac{d}{c})$. The volume of this tetrahedron is $\frac{d^3}{6abc}$. In the present problem this equals 6 because $a = 1, b = 2, c = 3$

and $d = 6$. So the volume of the entire given solid is 8 times this, i.e. 48. ■

An easy problem once the idea of an analogous problem in the plane strikes.

Q.10 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $\frac{d^2 f(x)}{dx^2}$ is positive for all $x \in \mathbb{R}$ and suppose $f(0) = 1, f(1) = 4$. Which of the following is not a possible value of $f(2)$?

- (A) 7 (B) 8 (C) 9 (D) 10

Answer and Comments: (A). As $f''(x) > 0$ everywhere, the graph of $y = f(x)$ is strictly concave upwards. That means that for any points $A = (a, f(a))$ and $B = (b, f(b))$ on it, the chord AB lies above the graph of $f(x)$ for $a \leq x \leq b$. In particular, if we take $x = \frac{a+b}{2}$, then

$$f\left(\frac{a+b}{2}\right) < \frac{f(a) + f(b)}{2} \quad (1)$$

In the present problem, taking $a = 0, b = 2$ we have

$$4 = f(1) < \frac{f(0) + f(2)}{2} = \frac{1 + f(2)}{2} \quad (2)$$

which gives $8 < 1 + f(2)$, i.e. $f(2) > 7$. So 7 cannot be a possible value of $f(2)$. ■

A simple problem once the idea of concavity strikes. Those not familiar with it can get the answer directly by applying the LMVT. First, applying it to the intervals $[0, 1]$ and $[1, 2]$, there exist $c_1 \in (0, 1)$ and $c_2 \in (1, 2)$ such that

$$f'(c_1) = f(1) - f(0) = 3 \quad (3)$$

$$\text{and } f'(c_2) = f(2) - f(1) = f(2) - 4 \quad (4)$$

Now since $f''(x) > 0$ everywhere, $f'(x)$ is strictly increasing (again by the LMVT). Since $c_1 < 1 < c_2$, this means $f'(c_1) < f'(c_2)$. Combining this with the equations above, we get

$$3 < f(2) - 4 \quad (5)$$

which gives that $f(2) > 7$. (This is hardly a different proof because the proof that the chord lies above the graph is also obtained by two applications of LMVT. Once to f (twice) and then to f' .)

The inequality (1) is a special case of what is called the **Jensen's inequality**. See Theorem 6 in Chapter 13.

Q.11 Let $f(x) = e^{-|x|}$, $x \in \mathbb{R}$ and $g(\theta) = \int_{-1}^1 f\left(\frac{x}{\theta}\right) dx$, $\theta \neq 0$.

Then $\lim_{\theta \rightarrow 0} \frac{g(\theta)}{\theta}$

- (A) equals 0 (B) equals $+\infty$
 (C) equals 2 (D) does not exist.

Answer and Comments: (D). Note that $f(x) = e^{-|x|}$ is an even function of x . To evaluate $g(\theta)$ for $\theta \neq 0$, we make a substitution

$$x = \theta u \quad (1)$$

Then

$$g(\theta) = \int_{-1}^1 f\left(\frac{x}{\theta}\right) dx = \int_{-1/\theta}^{1/\theta} f(u) \theta du \quad (2)$$

Hence

$$\frac{g(\theta)}{\theta} = \int_{-1/\theta}^{1/\theta} e^{-|u|} du \quad (3)$$

$$= 2 \int_0^{1/\theta} e^{-|u|} du \quad (4)$$

since the integrand is an even function of u and the interval of integration is symmetric about 0. (Note that for $\theta < 0$, the lower limit of integration in (2) is higher than the upper one. But that makes no difference.)

We now make two cases depending upon whether $\theta > 0$ or $\theta < 0$. If $\theta > 0$, then u is non-negative in the interval of integration and so $|u| = u$, giving

$$\begin{aligned} \frac{g(\theta)}{\theta} &= 2 \int_0^{1/\theta} e^{-u} du \\ &= -2e^{-u} \Big|_0^{1/\theta} \\ &= 2(1 - e^{-1/\theta}) \end{aligned} \quad (5)$$

As $\theta \rightarrow 0^+$, the exponent $-1/\theta$ tends to $-\infty$ and hence $e^{-1/\theta}$ tends to 0. So

$$\lim_{\theta \rightarrow 0^+} \frac{g(\theta)}{\theta} = 2 \quad (6)$$

On the other hand, when $\theta < 0$, $u \leq 0$ at all points of the interval of integration and so $-|u| = u$ for all u which gives

$$\frac{g(\theta)}{\theta} = 2 \int_0^{1/\theta} e^u du$$

$$\begin{aligned}
&= 2e^u \Big|_0^{1/\theta} \\
&= 2e^{1/\theta} - 2
\end{aligned} \tag{7}$$

This time too, as $\theta \rightarrow 0^-$, $1/\theta$ tends to $-\infty$ and so

$$\lim_{\theta \rightarrow 0^-} \frac{g(\theta)}{\theta} = -2 \tag{8}$$

From (6) and (8) we conclude that $\lim_{\theta \rightarrow 0} \frac{g(\theta)}{\theta}$ does not exist since the right and left handed limits are not equal. ■

The integration part is elementary. The problem is more a test of the ability to carefully consider the two cases that arise because of the ambiguity in the modulus function. Ironically, some candidates may mark the correct answer from (3) on the superficial ground that $\lim_{\theta \rightarrow 0} \frac{1}{\theta}$ does not exist. Apparently, option (B) is given to tempt such candidates. But these days it is often hammered on students that ∞ is a non-entity and therefore to say that some limit is infinite really says that it does not exist. Candidates who reach (4) but later fail to consider the two cases will mark (C) as the correct option.

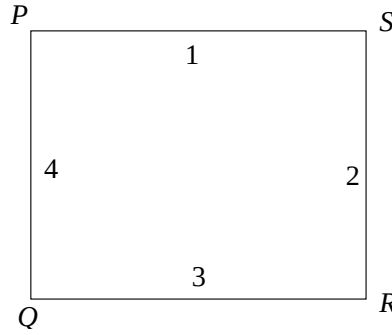
Q.12 The number of different ways to colour the vertices of a square $PQRS$ using one or more colours from the set {Red, Blue, Green, Yellow} such that no two adjacent vertices have the same colour is

(A) 36 (B) 48 (C) 72 (D) 84

Answer and Comments: (D). A typical problem on the application of the **principle of inclusion and exclusion**. Exercise (1.44) is very similar except that instead of a square, we have to paint the walls of a room and instead of 4 there are only 3 colours. Note that the fact that $PQRS$ is a square has little bearing on the problem. It could as well be any quadrilateral. Because of the high degree of symmetry in a square, two colourings which are different on the nose may be the same if one can be obtained from the other by a rotation, a flip or a reflection (either into a diagonal or into a line joining the midpoints of a pair of opposite sides). But in that case the problem will be of a very different class. (See Comment No.8 and Exercise (1.19) of Chapter 1 for examples of these type of problems.)

As the problem stands, there are in all 4^4 ways to assign the given colours to the four vertices P, Q, R, S of the square. This is because every colouring

is a function from the set $\{P, Q, R, S\}$ to the set of the given four colors. Call the set of all such functions as F . We want the subset, say G , of those functions f which take different values at every adjacent pair of vertices. In all there are four pairs of adjacent vertices, corresponding to the sides of the square. We number them as 1,2,3 and 4 as shown in the accompanying diagram.



For $i = 1, 2, 3, 4$, let F_i be the set of those functions in which the end points of the i -th side have the same colour, there being no restriction on the colours of the other two vertices. Then clearly for every i ,

$$|F_i| = 4^3 \quad (1)$$

because we may think of the end points of the side i as fused together into a single vertex. We are then effectively colouring only 3 vertices with four colours.

We now consider intersections of the form $F_i \cap F_j$ where $i \neq j$. There are $\binom{4}{2} = 6$ such intersections. If i, j are adjacent sides, then three vertices must get the same colour and the fourth one any colour. If i and j are parallel sides then also the two end points of i must have the same colour and the remaining two the same colour which may be different from that of the end points of i . So, in any case, for all $i \neq j$, we have

$$|F_i \cap F_j| = 4^2 \quad (2)$$

With 3 or 4 distinct sides, all four vertices are covered by their endpoints and they must get the same colour which can be any of the given four colours. So

$$|F_i \cap F_j \cap F_k| = 4 \quad (3)$$

for any three distinct i, j, k . There are four such triplet. And finally,

$$|F_1 \cap F_2 \cap F_3 \cap F_4| = 4 \quad (4)$$

We now apply the principle of inclusion and exclusion to get

$$\begin{aligned}
 |G| &= |F| - \sum_{1 \leq i \leq 4} |F_i| + \sum_{1 \leq i < j \leq 4} |F_i \cap F_j| - \sum_{1 \leq i < j < k \leq 4} |F_i \cap F_j \cap F_k| \\
 &\quad + |F_1 \cap F_2 \cap F_3 \cap F_4|
 \end{aligned} \tag{5}$$

$$\begin{aligned}
 &= 4^4 - 4 \times 4^3 + 6 \times 4^2 - 4 \times 4^1 + 4^1 \\
 &= 256 - 256 + 96 - 16 + 4 \\
 &= 84
 \end{aligned} \tag{6}$$

Hence the desired number of colourings is 84. ■

A routine problem for those familiar with the principle of inclusion and exclusion. The calculations are somewhat repetitious and prone to numerical slips. But as there is no need to explain the reasoning, the problem is a reasonable one as an MCQ.

Q.13 Define $a = p^3 + p^2 + p + 11$ and $b = p^2 + 1$ where p is any prime number. Let $d = \gcd(a, b)$. Then the set of possible values of d is

- (A) 1,2,5 (B) 2,5,10 (C) 1,5,10 (D) 1,2,10.

Answer and Comments: (B). Since d divides both a and b , it divides any integral linear combination of them. In particular it divides

$$a - (p + 1)b \tag{1}$$

which is simply 10. The only possible positive integers that divide 10 are 1, 2, 5 and 10. However, d cannot be 1 because in that case a and b would be relatively prime. But that can be dismissed by observing that if p is an odd prime, then both a and b are even, while if $p = 2$, the only even prime, then $a = 25$ and $b = 5$ which are not relatively prime either. So if one of the options is correct, it has to be (B). For the sake of completeness, we have to show that each of 2, 5 and 10 occurs as the g.c.d. of a and b for some prime p . We already saw that this is the case for $d = 5$. If we take $p = 3$, then $a = 50$ and $b = 10$ which have 10 as their g.c.d.. Finally, if we take $p = 5$, then $a = 166$ and $b = 26$ which have only 2 as their common factor and hence their g.c.d. So the possible values of d are 2, 5 and 10. ■

A simple problem about greatest common divisors. The linear combination appearing in (1) may appear tricky. But it is based on the method of finding the g.c.d. of two numbers we learn in schools. Finding the prime factorisation of the two numbers will give the g.c.d. but is often very time consuming. Instead, divide the larger number by the smaller and let r be the remainder. Then divide b by r and let s be the remainder. Continue this process till the remainder is 0. Then the last but one remainder is

the g.c.d. For example the g.c.d. of 1092 and 195 can be found from

$$\begin{aligned} 1092 &= 5 \times 195 + 117 \\ 195 &= 1 \times 117 + 78 \\ 117 &= 1 \times 78 + 39 \\ 78 &= 2 \times 39 + 0 \end{aligned}$$

Thus 39 is the g.c.d. of 1092 and 195.

The same procedure can be adopted for finding the g.c.d. of two polynomials, except that we now go by the degrees of the two polynomials. That is, we divide the polynomial of higher degree by that of a lower one to get a remainder which is a polynomial of a lower degree than both and repeat the process till we reach a polynomial of degree 0. If we adopt this procedure to the two polynomials $p^3 + p^2 + p + 11$ and $p^2 + 1$, we get

$$p^3 + p^2 + p + 11 = (p^2 + 1)(p + 1) + 10 \quad (2)$$

We are lucky because already we have reached a polynomial of degree 0 as the remainder. We can now express 10 as a linear combination of the polynomials $p^3 + p^2 + p + 11$ and $p^2 + 1$ by working backwards. That is,

$$10 = p^3 + p^2 + p + 11 - (p + 1)(p^2 + 1) \quad (3)$$

which is precisely (1).

This method of finding the g.c.d. is called the **euclidean algorithm**. (Nowadays, the term is used more for the crucial step in the algorithm, viz. given a and b , expressing a in the form $bq + r$ where the remainder r is less than the divisor b in some sense.) The problem makes an imaginative use of it.

Q.14 Consider all 2×2 matrices whose entries are distinct and taken from the set $\{1, 2, 3, 4\}$. The sum of the determinants of all such matrices is

- (A) 24 (B) 10 (C) 12 (D) 0

Answer and Comments: (D). Every matrix of the given type corresponds to a permutation of the four symbols 1,2,3 and 4. (Normally, in a permutation, the entries are written in a row. But that is a superficial difference. Given an $m \times n$ matrix A , we can write its rows out one after the other in a single row of length mn . Conversely, from any such row we can recover an $m \times n$ matrix by merely breaking it into m subrows of length n and writing them below one another.)

So, in all there are $4! = 24$ 2×2 matrices that can be formed from the four symbols. It would be horrendous to write them all, calculate the determinant of each and add. Instead, we group them together using interchanges of rows and/or columns and add the determinant of all matrices in the same group.

Thus if we start with a matrix $A_1 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and interchange its rows, we get the matrix $A_2 = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$. There is no point in interchanging the rows again, because that will give back the matrix A_1 . But we can interchange the columns of A_2 to get a matrix $A_3 = \begin{bmatrix} d & c \\ b & a \end{bmatrix}$. Again no point in interchanging columns at this stage. But we can interchange the rows to get $A_4 = \begin{bmatrix} c & a \\ d & b \end{bmatrix}$. Note that all these four matrices are distinct. But now no matter whether we interchange rows or columns we do not get a new matrix. So these four matrices form a set of matrices which is closed under interchange of rows as well as of columns. The determinants of these four matrices cancel and add to 0. So the set of 24 possible matrices is decomposed into six disjoint subsets of four matrices each, whose determinants add to 0. So the determinants of all 24 matrices add to 0. ■

A simple but imaginatively designed problem requiring nothing about determinants beyond the effect of an interchange of rows or columns.

- Q.15 Let a, b, c and d be four non-negative real numbers where $a + b + c + d = 1$. The number of different ways one can choose these numbers such that $a^2 + b^2 + c^2 + d^2 = \max\{a, b, c, d\}$ is ...

(A) 1 (B) 5 (C) 11 (D) 15

Answer and Comments: (D). Assume, without loss of generality that d is the maximum out of a, b, c, d . Then we have a system of two equations:

$$a + b + c + d = 1 \quad (1)$$

$$\text{and } a^2 + b^2 + c^2 + d^2 = d \quad (2)$$

This is a system of two equations in four unknowns. In general it will have infinitely many solutions. If the second equation, like the first one, were linear, there is a systematic method to display the totality of solutions. But for a non-linear system there is no such short cut. The solution set of (1) is a 3-dimensional hyperspace in the space $\mathbb{R}^4$ while that of (2) is the sphere in $\mathbb{R}^4$ of radius $\frac{1}{2}$ and centred at $(0, 0, 0, \frac{1}{2})$. If we had two or three variables instead of 4, we could identify these sets as some familiar ones. But in $\mathbb{R}^4$ we cannot even visualise them.

Secondly, even if we were to somehow identify the solution set of the system, we are interested only in those members (a, b, c, d) of it in which a, b, c, d are all non-negative and further a, b, c all lie in $[0, d]$. This will be a horrendous task which can hardly be expected to be done in a couple of minutes.

Fortunately, there is an easier way where we apply the data and the inequalities simultaneously. As before, let d be the largest of a, b, c, d . Then we have

$$\begin{aligned} d &= d(a + b + c + d) \\ &= ad + bd + cd + d^2 \end{aligned} \tag{3}$$

$$\geq a^2 + b^2 + c^2 + d^2 \tag{4}$$

$$= d \tag{5}$$

where in going from (3) to (4) we have actually applied three inequalities, viz. $ad \geq a^2, bd \geq b^2$ and $cd \geq c^2$. Thus we started with d and after applying the three inequalities, we finally landed back at d . So none of these three inequalities is strict. They are all equalities. In particular,

$$ad = a^2 \tag{6}$$

This can happen only when $a = 0$ or $a = d$. The same applies to b and c . So, out of the numbers a, b, c those that are not equal to d are zero. Hence if we write the numbers a, b, c, d in an ascending order, then some top k of them equal d and the remaining are 0. Here k can vary from 1 to 4. Their sum $a + b + c + d$ is simply kd . Since $kd = 1$, we have four cases:

- (i) $k = 1, d = 1$. Then a, b, c are all 0.
- (ii) $k = 2$. Then $a = b = 0$ and $c = d = 1/2$. (Note that the squares of a, b, c and d do add to $\frac{1}{2}$ which is indeed d .)
- (iii) $k = 3$. Here $a = 0$ and $b = c = d = \frac{1}{3}$.
- (iv) $k = 4$. Here $a = b = c = d = \frac{1}{4}$.

The choice of d as the maximum of a, b, c, d was arbitrary. If we change it, we shall get more solutions. In (i), there will be 3 more solutions. In (ii), exactly two numbers out of four equal $\frac{1}{2}$ each and the remaining 0. So there are $\binom{4}{2} = 6$ solutions of this type. In (iii) there is only one zero entry, the remaining are $\frac{1}{3}$ each. There are 4 such solutions. Finally, there is only one solution of type (iv).

Hence the total number of possible solutions is $4 + 6 + 4 + 1 = 15$. ■

The logic can be easily extended to the case of n non-negative real numbers, say $a_1, a_2, \dots, a_n$ which add to 1 and whose squares add to $\max\{a_1, a_2, \dots, a_n\}$. In any such collection, all non-zero numbers are equal and add to 1. If k is the number of such non-zero numbers out of $\{a_1, a_2, \dots, a_n\}$, then there are $\binom{n}{k}$ collections of this type. Hence the total number of such collections is $\sum_{k=1}^n \binom{n}{k}$. By the binomial theorem, (or by counting the number of non-empty subsets of a set with n elements),

this equals $2^n - 1$. In the present problem, $n = 4$ and so the answer is $2^4 - 1 = 15$.

To appreciate the incisive nature of this argument, we work out the cases $n = 2$ and $n = 3$ by the algebraic approach, i.e. by solving a system of two equations and then taking only those where one of the variables assumes values higher than the others.

When we have only two variables a and b the corresponding system is

$$a + b = 1 \quad (7)$$

$$\text{and } a^2 + (b - \frac{1}{2})^2 = \frac{1}{4} \quad (8)$$

Here the solution set is the intersection of a circle with a line. It is the boundary of a chord with end points $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$. Both are solutions of the problem. In addition, if we assume $a \geq b$, we have $(1, 0)$ as one more solution. Hence there are 3 solutions in all.

With 3 variables, the system is

$$a + b + c = 1 \quad (9)$$

$$\text{and } a^2 + b^2 + (c - \frac{1}{2})^2 = \frac{1}{4} \quad (10)$$

The solution set is the intersection of a sphere and a plane. So it is a circle, say C . Even if we take only the part of it in the first orthant, it has infinitely many points. We can explicitly identify them introducing what are called spherical polar coordinates with the pole at the centre of the sphere, viz. at $(0, 0, \frac{1}{2})$. As all points are at the same distance from the pole, the parametrisation of the sphere can be done with two parameters θ and ϕ .

$$a = \frac{1}{2} \sin \phi \cos \theta \quad (11)$$

$$b = \frac{1}{2} \sin \phi \sin \theta \quad (12)$$

$$c = \frac{1}{2} + \frac{1}{2} \cos \phi \quad (13)$$

where $\theta \in [0, 2\pi]$ and $\phi \in [0, \pi]$.

The requirement $a + b + c = 1$ translates into

$$\sin \phi \cos \theta + \sin \phi \sin \theta = 1 - \cos \phi \quad (14)$$

Simplification gives either $\sin \frac{\phi}{2} = 0$ (which happens only at the north pole $(0, 0, 1)$) or

$$\tan \frac{\phi}{2} = \sin \theta + \cos \theta \quad (15)$$

We are interested only in the solutions in the first orthant. This restricts θ to $[0, \frac{\pi}{2}]$. Even then there are infinitely many solutions. But we want only those in which a, b are both less than or equal to c . The requirement $b \leq c$ reduces to $\sin \phi \cos \theta \leq 1 + \cos \phi$ and, further to $\tan \frac{\phi}{2} \cos \theta \leq 1$ since $\cos \frac{\phi}{2} \neq 0$ as otherwise ϕ would be π and c would be 0. Substitution from (15) gives

$$(\sin \theta + \cos \theta) \cos \theta \leq 1 \quad (16)$$

which simplifies to

$$\sin \theta \cos \theta \leq \sin^2 \theta \quad (17)$$

One possibility is that $\sin \theta = 0$ which gives $b = 0$ and further $\cos \theta = 1$, $\phi = \frac{\pi}{2}$ whence $a = c = \frac{1}{2}$. So, $(\frac{1}{2}, 0, \frac{1}{2})$ is a solution. If $\sin \theta \neq 0$, then $\sin \theta > 0$ as $\theta \in [0, \frac{\pi}{2}]$ and (17) reduces to

$$\cos \theta \leq \sin \theta \quad (18)$$

Similarly, the requirement $a \leq c$ leads to two possibilities : either the solution $(0, \frac{1}{2}, \frac{1}{2})$ or the inequality

$$\sin \theta \leq \cos \theta \quad (19)$$

So, when both a and b are less than or equal to c , (a, b, c) must be $(0, 0, 1)$, $(\frac{1}{2}, 0, \frac{1}{2})$ or $(0, \frac{1}{2}, \frac{1}{2})$ or else (18) and (19) must both hold. But that means $\theta = \frac{\pi}{4}$. In this case (15) gives $\tan \frac{\phi}{2} = \sqrt{2}$ whence $\sin \phi = \frac{2\sqrt{2}}{3}$ and $\cos \phi = \frac{1-2}{3} = -\frac{1}{3}$. Substitution into (11), (12) and (13) gives $(a, b, c) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

Summing up, we have four solutions in which c is the maximum of a, b, c . In all these, the non-zero entries are all equal. Permuting the variables, we get $(1, 0, 0)$, $(0, 1, 0)$ and $(\frac{1}{2}, \frac{1}{2}, 0)$ as three more solutions. So in all there are 7 solutions which tallies with the earlier count, viz. $2^3 - 1$.

This case alone is sufficient to illustrate the superiority of the earlier method. The technique adopted in it may be called **proving equality through a cycle of inequalities**. By repeated applications of the transitivity property of the order relation (that is, $x \geq y \geq z$ implies $x \geq z$), we get that if

$$x_1 \geq x_2 \geq x_3 \geq \dots \geq x_{n-1} \geq x_n \quad (20)$$

then $x_1 \geq x_n$. This is sheer common sense and we often resort to it without even consciously applying transitivity (much the same way that we manipulate many expressions with repeated applications of the commutativity and associativity of addition, without ever mentioning this). It is like seating n persons in a row. If every person (except the last one) in the row is at least as tall as his immediate right neighbour, then surely

the leftmost person is the tallest and the rightmost person the shortest. But what if these persons are seated around a circle? Then the only way everyone will be at least as tall as his rightmost neighbour is when all of them have exactly the same height. Mathematically, if we have

$$x_1 \geq x_2 \geq x_3 \geq \dots \geq x_{n-1} \geq x_n \geq x_1 \quad (21)$$

then equality holds throughout, i.e.

$$x_1 = x_2 = \dots = x_n \quad (22)$$

In a way we have used this technique of cyclic inequalities when we used (18) and (19) together to conclude that $\sin \theta = \cos \theta$. But here the cycle is only of length 2 and nobody calls it a cycle. Rather, it is simply called the **anti-symmetry property** of the order relation, i.e. $x \leq y$ and $y \leq x$ together imply $x = y$. Actually, it is transitivity and anti-symmetry together which constitute a mathematical proof that (21) implies (22).

The technique of proving equality through a cycle of inequalities is rarely used with an explicit mention. It is there indirectly in the Sandwich Theorem for limits, especially when the number of intermediate sequences (or functions) is more than two. It is rarely applied where limits are not involved. (For those familiar with linear algebra, it is used in the proof that a matrix is diagonalisable if and only if for every eigenvalue of it, its geometric and algebraic multiplicities are equal. Without going into the technicalities, the underlying idea is merely that if there are several boxes each containing at least as many green balls as red ones and the total number of green balls in the boxes is the same as the total number of red balls, then every box contains as many green balls as red ones.) Therefore the solution to the present problem is exceptional in this sense. Such exceptional methods generally apply to problems that are themselves special rather than generic. For example, it would fail if $a^2 + b^2 + c^2 + d^2$ were given to be some other multiple of the maximum of a, b, c, d . In this sense the problem is what may be called accidentally solvable. (See the comments on Problem 41 and Problem 58 in Paper 1 of JEE 2013.)

Q.16 The polynomial $x^4 + 4x + c = 0$ has at least one real root if and only if

$$(A) \ c < 2 \quad (B) \ c \leq 2 \quad (C) \ c < 3 \quad (D) \ c \leq 3$$

Answer and Comments: (D). A better wording would have been ‘polynomial equation’ rather than ‘polynomial’. In fact, the problem has little to do with polynomials. Let

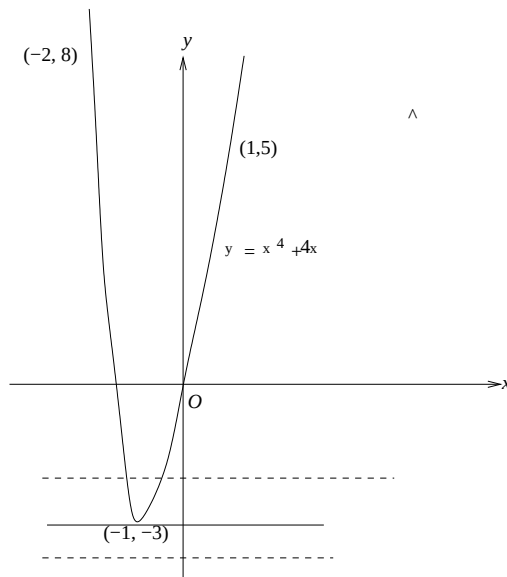
$$f(x) = x^4 + 4x \quad (1)$$

If x_0 is any solution of $f(x) + c = 0$, that means the line $y = -c$ cuts the graph of $y = f(x)$ at the point $(x_0, -c)$. We have to find the largest value

of c for which this is possible. The easiest approach is to draw the graph of $y = x^4 + 4x$. Clearly $f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$. So $f(x)$ has a minimum at some point. To find it, we note that

$$f'(x) = 4x^3 + 4x = 4(x^3 + 1) \quad (2)$$

vanishes only at $x = -1$. $f(-1) = -3$. $f(x)$ is decreasing on $(-\infty, -1)$ and increasing on $(-1, \infty)$. So it has a (unique) minimum at -1 . The line $y = -3$ touches the graph at $(-1, -3)$.



For $h < -3$ the line $y = h$ does not intersect the graph (see the lower dotted line) while for $h > -3$ the line $y = h$ meets it in more than one points (see the upper dotted line).

Now to say that $f(x) + c = 0$ has at least one solution is equivalent to saying that the line $y = -c$ meets the graph $y = f(x)$ at at least one point. By what we just showed, this happens if and only if $-c \geq -3$, i.e. $c \leq 3$. ■

A simple problem once the idea of relating a solution of $f(x) + c = 0$ with the point of intersection of the line $y = -c$ with the graph of $y = f(x)$ strikes. After that it is a very simple problem of finding the minimum of the function $f(x) = x^4 + 4x$. As said earlier, polynomials have little to do with the problem.

Q.17 The number of all integer solutions of the equation $x^2 + y^2 + x - y = 2021$ is

- (A) 5 (B) 7 (C) 1 (D) 0

Answer and Comments: (D). The numbers $x^2 + x$ and $y^2 - y$ are even for any x, y , each of them being the product of two consecutive integers. So the L.H.S. is even while the R.H.S. is odd. Hence there cannot be any solution. ■

A highly disappointing problem, especially so after Problem 15. It would have been an interesting problem if, like Problem 15, there were some solutions and some way to identify them. Or at least the non-existence of solutions should have come after some non-trivial reasoning.

- Q.18 The number of different values of a for which the equation $x^3 - x + a = 0$ has two identical real roots is

(A) 0 (B) 1 (C) 2 (D) 3

Answer and Comments: (C). This problem is of the same spirit as Problem 16. In general, a horizontal line will cut the graph of a cubic in 1 or 3 points. But if it touches the graph, then two of these points coincide. This happens at those points where there is a local extremum. Taking $f(x) = x^3 - x$ we see that $f'(x) = 3x^2 - 1$ and so there is a local maximum at $x = -\frac{1}{\sqrt{3}}$ and a local minimum at $x = \frac{1}{\sqrt{3}}$. Hence there will be two horizontal lines which will meet the graph at two distinct points. Put differently, there will be two values of a for which $f(x) = -a$ has two distinct solutions. ■

The problem does not ask to identify these two values. They are simply the negatives of $f(\pm\frac{1}{\sqrt{3}})$.

There is little point in asking two problems with the same idea.

- Q.19 Suppose $f(x)$ is a twice differentiable function on $[a, b]$ such that $f(a) = 0 = f(b)$ and

$$x^2 \frac{d^2 f(x)}{dx^2} + 4x \frac{df(x)}{dx} + 2f(x) > 0$$

for all $x \in (a, b)$. Then

- (A) f is negative for all $x \in (a, b)$
 (B) f is positive for all $x \in (a, b)$
 (C) $f(x) = 0$ for exactly one $x \in (a, b)$
 (D) $f(x) = 0$ for at least two $x \in (a, b)$

Answer and Comments: (A). Note that in the given condition, viz.

$$x^2 \frac{d^2 f(x)}{dx^2} + 4x \frac{df(x)}{dx} + 2f(x) > 0 \tag{1}$$

for all $x \in (a, b)$, the last sign is of inequality and not of equality. Had it been so, it would have been a differential equation and there are methods available for solving it. But the solution would be of little use because the solution of a d.e. does not say much about what would happen if inequality holds. For example, the general solution of the d.e.

$$\frac{d^2 y}{dx^2} + y = 0 \quad (2)$$

is

$$y = A \cos x + B \sin x \quad (3)$$

where A and B are arbitrary constants whose values are to be determined from some initial conditions. But if we consider the inequality

$$\frac{d^2 y}{dx^2} + y > 0 \quad (4)$$

there are many functions which satisfy it which have nothing to do with the solutions (3) of (2). Indeed $y = x^m$ where m is any even positive integer satisfies (4) for all $x \neq 0$.

So there is no point in solving (1). We have to look for other ways. The positivity of the derivative or of the second derivative implies something about a function. For example, the positivity of the first derivative implies that the function is strictly increasing, positivity of the second derivative implies that it is strictly concave upward. But here we do not have a single derivative but a combination of the first and the second derivative (and the function itself). From this it cannot be concluded that either the first or the second derivative is positive throughout the given interval $[a, b]$.

If we could recast the L.H.S. of (1) as the second derivative of some new function $g(x)$ related to $f(x)$, then we can hope to get some information about $g(x)$ and from that some information about $f(x)$ (which is our goal).

Once this point is understood, the function $g(x)$ can be found with a little trial. There is little point in taking $g(x)$ as $h(x) + f(x)$ for some function $h(x)$ of our choice. In doing so, the derivatives of $g(x)$ would simply fall apart into those of $f(x)$ and $h(x)$. So we try $g(x)$ be of the form $h(x)f(x)$ for some function $h(x)$. In that case the second derivative of $g(x)$ would have a term $h(x)f''(x)$. The first term of the L.H.S. of (1) is $x^2 f''(x)$. This suggests that our best bet is to take $h(x)$ as x^2 . We indeed do so and let

$$g(x) = x^2 f(x) \quad (5)$$

A direct calculation then gives $g'(x) = x^2 f'(x) + 2xf(x)$ and $g''(x) = x^2 f''(x) + 4xf'(x) + 2f(x)$. We are lucky that this is precisely the L.H.S. of (1). So, without any further manipulations, (1) can now be recast as

$$g''(x) > 0 \quad (6)$$

for all $x \in (a, b)$. This is precisely the condition for (strict) concavity upward of $g(x)$ over $[a, b]$. Note that since f vanishes at both a and b , so does g . So the chord joining the points $(a, g(a))$ and $(b, g(b))$ is simply along the x -axis. For any $c \in (a, b)$, the point $(c, g(c))$ lies below this chord. That means $c^2 f(c) < 0$ whence $f(c) < 0$ too. So (A) holds. ■

An excellent problem which penalises those who know more but don't know where *not* to use their knowledge. The differential equation

$$x^2 \frac{d^2 f(x)}{dx^2} + 4x \frac{df(x)}{dx} + 2f(x) = 0 \quad (7)$$

is an example of what are called Cauchy-Euler types of equations. The coefficients of the derivatives are not constants. But if we change the independent variable to $z = \ln x$, then the resulting d.e. in y and z is a second order linear d.e. with constant coefficients and can be solved by finding its characteristic roots. All this work can be shortened by trying a solution of the form $f(x) = x^m$ for some integer m . Then the equation reduces to

$$m(m-1)x^m + 4mx^m + 2x^m = 0 \quad (8)$$

which gives a quadratic for m , viz. $m^2 + 3m + 2 = 0$. Its roots are $m = -1$ and $m = -2$. So the general solution of (7) is

$$f(x) = \frac{A}{x} + \frac{B}{x^2} \quad (9)$$

where A and B are some constants. The initial conditions $f(a) = 0 = f(b)$ give $A = 0 = B$. So $f(x)$ is identically 0. This is puzzling but also an eye opener that nothing is to be gained by solving the d.e. associated with (1). Some other approach has to be tried.

For those who do not know differential equations, ignorance is bliss in this problem.

Q.20 Consider the following two subsets of $\mathcal{C}$.

$$A = \left\{ \frac{1}{z} : |z| = 2 \right\} \text{ and } B = \left\{ \frac{1}{z} : |z - 1| = 2 \right\}$$

Then

- (A) A is a circle but B is not a circle
- (B) B is a circle but A is not a circle
- (C) A and B are both circles
- (D) Neither A nor B is a circle

Answer and Comments: (C). Both the sets consist of reciprocals of points on a circle. But in A , since $|\frac{1}{z}| = \frac{1}{|z|}$, A is nothing but $\{z : |z| = \frac{1}{2}\}$ which is clearly a circle. In B , however, although all points of the starting circle are equidistant from 1, their reciprocals need not be equidistant from the reciprocal of the centre, because reciprocation does not preserve distance. In fact, they need not be equidistant from *any* (fixed) complex number. To see what the set B really looks like, suppose $z = x + iy$ is a typical point of B . Then $\frac{1}{z}$ is a point at a distance 2 from 1. So, we have

$$|\frac{1}{z} - 1| = 2 \quad (10)$$

It is easier to work with squares of the distances. So we rewrite this as

$$|\frac{1}{z} - 1|^2 = 4 \quad (11)$$

To get an idea of what type of curve this is, we identify the real and imaginary parts of $\frac{1}{z}$. There is a standard way of doing this.

$$\frac{1}{z} = \frac{1}{x + iy} \quad (12)$$

$$= \frac{x - iy}{x^2 + y^2} \quad (13)$$

$$= \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2} \quad (14)$$

Hence (4) translates into

$$(\frac{x}{x^2 + y^2} - 1)^2 + (\frac{y}{x^2 + y^2})^2 = 4 \quad (15)$$

which simplifies to

$$\frac{x^2 + y^2}{(x^2 + y^2)^2} - \frac{2x}{x^2 + y^2} = 3 \quad (16)$$

and further, after multiplying throughout by $x^2 + y^2$ to

$$1 - 2x = 3(x^2 + y^2) \quad (17)$$

No further simplification is needed since already this is a quadratic with no xy term and the coefficients of x^2 and y^2 are equal. Hence both A and B are circles. ■

A simple but interesting problem. It is very easy to see that A is a circle. Since the reasoning for A does not apply to B , casual candidates will mark (A) as the option. But those who work it out as above will realise that B is also a circle. (Although not asked, B has radius $\frac{2}{3}$ and centre at $-\frac{1}{3}$. Neither can be predicted readily from the radius and the centre of the circle $\{z : |z - 1| = 2\}$.)

Q.21 For a positive integer n the equation

$$x^2 + n = y^2$$

x, y integers, does not have a solution if and only if

- (A) $n = 2$
- (B) n is a prime number
- (C) n is an odd number
- (D) n is an even number not divisible by 4

Answer and Comments: (D). The given equation can be written as

$$(y - x)(y + x) = n \quad (1)$$

No matter what the integers x and y are, $y + x$ and $y - x$ have the same parity. That is, either both are even or both odd. In the first case n is even and, in fact, divisible by 4. In the second it is odd. Conversely, when n is odd we can easily factorise it in the form (1) by taking $y - x = 1$ and $y + x = n$ which has a solution, viz. $y = \frac{n+1}{2}$ and $x = \frac{n-1}{2}$. Also when n is even and divisible by 4, say $n = 4m$, we can take $y = m + 1$ and $x = m - 1$ and (1) is satisfied.

Therefore a necessary and sufficient condition for n to be expressed in the form (1) is that either it is odd or divisible by 4. Hence a necessary and sufficient condition that n cannot be expressed as (1) is that it is even but not divisible by 4. ■

Another simple problem involving only parity of integers. But this one is not as easy as Problem 17.

Q.22 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be any twice differentiable function such that its second derivative is continuous and $\frac{df(x)}{dx} \neq 0$ for all $x \neq 0$.

If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \pi$, then

- (A) for all $x \neq 0$, $f(x) > f(0)$
- (B) for all $x \neq 0$, $f(x) < f(0)$
- (C) for all x , $\frac{d^2f(x)}{dx^2} > 0$
- (D) for all x , $\frac{d^2f(x)}{dx^2} < 0$

Answer and Comments: (A). We are given that f' does not vanish except possibly at 0. Since f' is continuous (in fact, differentiable), it follows by IVP that f' maintains its sign on $(0, \infty)$ and also on $(-\infty, 0)$. (These two signs may be different or same.)

From the information $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \pi$, we can roughly conclude that near 0, $f(x)$ is comparable to πx^2 . Therefore, both $f(0)$ and $f'(0)$ must vanish while $f''(0) > 0$. Once this is granted, it follows that $f(x)$ has a local minimum at 0. But since $f''(0) > 0$ while $f'(0) = 0$, we get that for small values of $|x|$, $f'(x) > 0$ if $x > 0$ and $f'(x) < 0$ for $x < 0$. But since f' maintains its sign on $(-\infty, 0)$ and on $(0, \infty)$, we have that $f'(x) > 0$ for all $x > 0$ and $f'(x) < 0$ for all $x < 0$. Hence (using Lagrange MVT), f is strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0, \infty)$. Therefore $f(x)$ has a strict global minimum at 0. That is precisely (A).

For a rigorous proof that $f(0) = 0 = f'(0)$, we apply Taylor's theorem. As $f''(x)$ is given to exist everywhere, for every $x \neq 0$ we can write

$$f(x) = f(0) + f'(0)x + \frac{f''(c_x)}{2}x^2 \quad (1)$$

for some $c_x \in (0, x)$ (which may depend on x and with the understanding that $c_x \in (x, 0)$ in case $x < 0$). Hence for all $x \neq 0$ we have

$$\frac{f(x)}{x^2} = \frac{f(0)}{x^2} + \frac{f'(0)}{x} + \frac{f''(c_x)}{2} \quad (2)$$

As $x \rightarrow 0$, the L.H.S. is given to tend to a finite limit, viz. π . On the R.H.S., the last term tends to $f''(0)$ by continuity of f'' since as x tends to 0, so does c_x which lies in $(0, x)$.

So, unless both $f(0)$ and $f'(0)$ vanish, the first two terms on the R.H.S. have no limit and we get a contradiction. The fussy ones may still argue that it is sometimes possible for the sum of two functions to have a limit even when neither does, a trivial example being when the functions are $\frac{1}{x}$ and $\frac{-1}{x}$. To satisfy them, we divide (1) first by x rather than by x^2 to get that for all $x \neq 0$

$$\frac{f(x)}{x} = \frac{f(0)}{x} + f'(0) + \frac{f''(c_x)}{2}x \quad (3)$$

Now all except the first term on the R.H.S. tend to finite limits. (The L.H.S. and the last term on the R.H.S. actually tend to 0 since we can write the L.H.S. as $x \frac{f(x)}{x^2}$.) So, the first term of the R.H.S. tends to a finite limit and that can happen only when $f(0) = 0$. We can now go back to (2) and conclude that $f'(0) = 0$ too.

Hence the proof of (A) is complete. ■

For the sake of completeness, we must also rule out the other three options. The counterexample $f(x) = \pi x^2$ rules out (B) and (D) but not

(C) since here $f''(x) = 2\pi$ is positive at all points. We can modify it for (C). We note that in the proof above, we used the positivity of f'' only at 0. We would be through if f' is strictly increasing on $\mathbb{R}$ even if f'' vanishes at some points of $\mathbb{R}$. A clue in this direction is provided by Problem 7, where we showed that the function $\sin x + x$ is strictly increasing even though its derivative vanishes at some points. We integrate it from 0 to x to get $1 - \cos x + \frac{x^2}{2}$ as an example of a function whose derivative is strictly increasing even though the second derivative vanishes at some points. We multiply it by π to ensure that the hypothesis is satisfied. So, we let $f(x) = \pi(1 - \cos x + \frac{x^2}{2})$. This example meets all the parts of the hypotheses but $f''(x) = \pi(\cos x + 1)$ is not positive at some points.

This is a good problem on theoretical calculus, But its excellence is marred by the confusing data. First of all, continuity of f'' is needed only at 0. Secondly the number π has no role. It could have been any positive number. Perhaps it was the intention to test the ability to weed out the incidental part of the data and focus on the essentials, which in this problem, is to recognise that $f(0)$ and $f'(0)$ both vanish but $f''(0)$ is positive. Option (C) is probably given to tempt those who work with the example $f(x) = \pi x^2$. If, as in JEE, there is a possibility that more than one options are correct, this would indeed be a good trap. But since only one option is correct and (A) fits the table at least for the function $f(x) = \pi x^2$, a candidate need not bother to find a counterexample for (C).

Q.23 Let us denote the fractional part of a real number x by $\{x\}$. (Note $\{x\} = x - [x]$ where $[x]$ is the integer part of x .) Then

$$\lim_{n \rightarrow \infty} \{(3 + 2\sqrt{2})^n\}$$

- (A) equals 0
- (B) equals 1
- (C) equals $\frac{1}{2}$
- (D) does not exist

Answer and Comments: (B). Call $(3 + 2\sqrt{2})^n$ as a_n . By the binomial theorem

$$a_n = \sum_{r=0}^n \binom{n}{r} 3^{n-r} (2\sqrt{2})^r \quad (1)$$

The terms in which r is even are all integers and do not contribute anything to the fractional part of a_n . But those in which r is odd do because $(\sqrt{2})^r = 2^{(r-1)/2} \sqrt{2}$ is an irrational number.

It is not easy to calculate the fractional part of a_n directly. But if we let $b_n = (3 - 2\sqrt{2})^n$, then

$$b_n = \sum_{r=0}^n \binom{n}{r} 3^{n-r} (-2\sqrt{2})^r \quad (2)$$

The terms in (1) and (2) are the same except that those with r odd are the negatives of those in (1). So, if we add a_n and b_n these terms cancel and we get an integer, say c_n . Clearly, $a_n < c_n$ since $3 - 2\sqrt{2} > 0$ and therefore $b_n > 0$ for all n . But since $3 - 2\sqrt{2} < 1$, $0 < b_n < 1$ for all n . Hence c_n is an integer which is only slightly greater than a_n , the difference being less than 1. Therefore

$$[a_n] = c_n - 1 \quad (3)$$

and hence

$$\{a_n\} = a_n - [a_n] = a_n + 1 - c_n = 1 - b_n \quad (4)$$

Since $0 < 3 - 2\sqrt{2} < 1$, $b_n \rightarrow 0$ as $n \rightarrow \infty$. Therefore by (4), $\{a_n\} \rightarrow 1$ as $n \rightarrow \infty$. ■

A simple problem once the idea of introducing $3 - 2\sqrt{2}$, which is the conjugate quadratic surd of $3 + 2\sqrt{2}$, and adding the binomial expansions of a_n and b_n strikes. It is important that $0 < 3 - 2\sqrt{2} < 1$. The first inequality follows by squaring 3 and $2\sqrt{2}$. For the second, we rationalise $3 - 2\sqrt{2}$ as $\frac{1}{3+2\sqrt{2}}$ and observe that the denominator is bigger than 1. See Exercise (6.27) for more problems involving powers of quadratic surds.

Q.24 Let

$$p(x) = x^3 - 3x^2 + 2x, x \in \mathbb{R}$$

$$f_0(x) = \begin{cases} \int_0^x p(t)dt, & x \geq 0 \\ -\int_x^0 p(t)dt, & x < 0 \end{cases}$$

$$f_1(x) = e^{f_0(x)}, f_2(x) = e^{f_1(x)}, \dots, f_n(x) = e^{f_{n-1}(x)}$$

How many roots does the equation $\frac{df_n}{dx} = 0$ have in the interval $(-\infty, \infty)$?

- (A) 1 (B) 3 (C) $n + 3$ (D) $3n$

Answer and Comments: (B). The problem repeatedly constructs a new function $g(x)$ by taking the exponential of some old function $f(x)$ and then deals with this new function $g(x)$ (which will become the old function at the next stage). So, it is important to have a formula for the derivative

of a function so defined. The formula is an easy application of the chain rule. That is,

$$\frac{d}{dx}(e^{f(x)}) = e^{f(x)} f'(x) \quad (1)$$

Now the exponential function never vanishes. So from (1) we immediately get that if we start with $f_0(x)$ and recursively define

$$f_k(x) = e^{f_{k-1}(x)} \quad (2)$$

then $f'_k(x)$ has the same zeros as $f'_{k-1}(x)$. (This *does not* mean that $f'_k(x)$ and $f'_{k-1}(x)$ are the same functions. But they have the same zeros.)

Applying this fact repeatedly, it follows that $f'_n(x)$ has the same number of zeros as $f_0(x)$, which, of course, has to be found directly. The function $f_0(x)$ is given indirectly as the integral of some other function $p(x)$. We could evaluate $f_0(x)$ by integrating $p(t)$, i.e. by integrating $p(t)$ from 0 to x by first finding an antiderivative of $p(x)$. But that would be foolish because the Fundamental Theorem of Calculus tells us that

$$\frac{d}{dx} \left(\int_0^x p(t) dt \right) = p(x) \quad (3)$$

Hence

$$f'_0(x) = p(x) = x^3 - 3x^2 + x \quad (4)$$

By a direct factorisation, f'_0 has three roots, viz. 0, 1 and 2. By what we have already said, this is also the number of roots of each $f'_k(x)$. So the equation $\frac{df_n(x)}{dx} = 0$ has three roots, viz. 0, 1 and 2. ■

The solution uses the FTC which is non-trivial. But once you grant FTC, the problem itself is trivial, made to look non-trivial by unnecessarily introducing clumsy definitions. And, to add to the clumsiness, the definition of $f_0(x)$ is given in two separate parts, one for $x \geq 0$ and the other for $x < 0$. Although Riemann integrals are first defined over intervals of the form $[a, b]$, where $a < b$, by a standard convention, when $a > b$, the integral over $[a, b]$ is defined as the negative of the integral over $[b, a]$.

It is far from clear what is to be tested by such questions, perhaps only the ability to translate the data into simpler terms. To quote a classic joke, saying that a person's verticality changed to his horizontality simply means that he fell flat!

Q.25 For $0 \leq x < 2\pi$, the number of solutions of the equation

$$\sin^2 x + 2 \cos^2 x + 3 \sin x \cos x = 0$$

is

(A) 1 (B) 2 (C) 3 (D) 4

Answer and Comments: (D). We note that the product $\sin x \cos x$ can be expressed in terms of $\sin 2x$. If we could express $\sin^2 x + 2\cos^2 x$ in terms of $\cos 2x$, we would get a trigonometric equation in which some linear combination of $\sin 2x$ and $\cos 2x$ is equated with some number. For this, we multiply the given equation throughout by 2 to get

$$2 + 2\cos^2 x + 3\sin 2x = 0 \quad (1)$$

which can be recast as

$$3 + \cos 2x + 3\sin 2x = 0 \quad (2)$$

or

$$\cos 2x + 3\sin 2x = -3 \quad (3)$$

There is a standard procedure to solve such equations. Since the squares of the coefficients in the L.H.S. add to 10, we divide by $\sqrt{10}$ and get

$$\frac{1}{\sqrt{10}} \cos 2x + \frac{3}{\sqrt{10}} \sin 2x = -\frac{3}{\sqrt{10}} \quad (4)$$

We now introduce an acute angle α defined by

$$\sin \alpha = \frac{1}{\sqrt{10}} \quad \text{and} \quad \cos \alpha = \frac{3}{\sqrt{10}} \quad (5)$$

(Note that every angle in $[0, 2\pi)$ is determined uniquely by its sine and cosine.) (4) can now be written as

$$\sin(2x + \alpha) = -\cos \alpha = \sin(-\frac{\pi}{2} + \alpha) = \sin(\frac{3\pi}{2} + \alpha) \quad (6)$$

The general solution of (6) is

$$2x + \alpha = n\pi + (-1)^n(\frac{3\pi}{2} + \alpha) \quad (7)$$

where n is an integer. As α is an acute angle and $x \in [0, 2\pi)$, the L.H.S. lies in $[0, \frac{9\pi}{2})$. The R.H.S. lies in this range only for $n = 0, 2, 3$ and 5. Each of these values gives a solution. Hence there are four solutions to the given equation. (The problem does not ask to identify the solutions.)

For a simpler solution, we convert the equation to a quadratic equation in $\cos^2 x$ by squaring $3\sin x \cos x$. We begin by rewriting the given equation as

$$3\sin x \cos x = -1 - \cos^2 x \quad (8)$$

Squaring both the sides and rearranging the terms we get

$$10 \cos^4 x - 7 \cos^2 x + 1 = 0 \quad (9)$$

This is a quadratic in $\cos^2 x$ with roots $\frac{7 \pm \sqrt{49 - 40}}{20}$, i.e. $\frac{1}{2}$ and $\frac{1}{5}$. As both the roots are positive we get $\cos x = \pm \frac{1}{\sqrt{2}}$ and $\cos x = \pm \frac{1}{\sqrt{5}}$ as four possible values. For each possibility, there are two values of x in $[0, 2\pi)$. For example, if $\cos x = \frac{1}{\sqrt{2}}$, then x can be either $\frac{\pi}{4}$ or $\frac{7\pi}{4}$. Hence it is tempting to think that there are eight solutions. But the catch is that the value of $\cos x$ determines $\sin x$ only numerically, that is, the sign could be $+$ or $-$. It may happen that one of these two values satisfies the given equation but the second one does not. This happens because we have not solved the original equation but only the equation obtained by squaring, which is not a reversible operation. (Although $a = b$ implies $a^2 = b^2$, $a^2 = b^2$ implies only $a = \pm b$.) If the transformation from one equation to another is done by a reversible process (for example, merely recasting its terms or by a bijective substitution such as $x = 2y + 5$), then the solutions of the new equation will correspond to those of the original one.

To find out how many possible solutions we get from the four possible values of $\cos x$, viz. $\pm \frac{1}{\sqrt{2}}$ and $\pm \frac{1}{\sqrt{5}}$, we can analyse them one by one. In the original equation, $\sin x$ is uniquely determined by $\cos x$ as

$$\sin x = -\frac{1 + \cos^2 x}{3 \cos x} = -\frac{1}{3} \left(\cos x + \frac{1}{\cos x} \right) \quad (10)$$

If we now put $\cos x = \frac{1}{\sqrt{2}}$, we get

$$\sin x = -\frac{1}{3} \left(\frac{1}{\sqrt{2}} + \sqrt{2} \right) = -\frac{1}{\sqrt{2}} \quad (11)$$

Out of the two values of x in $[0, 2\pi)$ which satisfy $\cos x = \frac{1}{\sqrt{2}}$, viz. $\frac{\pi}{4}$ and $\frac{7\pi}{4}$, only the second one satisfies (11). Hence there is only one solution of the given equation in which $\cos x = \frac{1}{\sqrt{2}}$, viz. $x = \frac{7\pi}{4}$. The explanation is that when we try to determine $\sin x$ from $\cos x$ from the formula $\pm \sqrt{1 - \cos^2 x}$, there is an ambiguity. But now we are determining it by directly substituting $\cos x$ into the original equation. The other three possibilities also give rise to only one value each. There is no need to identify them. But now we know that the number of solutions is 4 and not 8.

The simplest solution (contributed by Siddhesh Naik), however, is to simply factorise the L.H.S. of the given equation as $(\sin x + \cos x)(\sin x + 2 \cos x)$. Each factor has two roots and they do not overlap! ■

A simple problem but a good test of the ability to choose the better method when several methods are available, and more importantly, to

exercise care when a particular method is adopted. The paper setters should have given 8 as one of the options to trap those who fail to exercise this care in the second method.

Q.26 Let $f : \mathbb{R} \rightarrow [0, \infty)$ be a continuous function such that

$$f(x+y) = f(x)f(y)$$

for all $x, y \in \mathbb{R}$. Suppose that f is differentiable at $x = 1$ and $\left. \frac{df(x)}{dx} \right|_{x=1} = 2$. Then the value of $f(1) \log_e f(1)$ is

- (A) e (B) 2 (C) $\log_e 2$ (D) 1

Answer and Comments: (B). The laws of indices imply that for any $a > 0$, the function

$$f(x) = a^x \tag{1}$$

satisfies the given relation, viz.

$$f(x+y) = f(x)f(y) \tag{2}$$

for all $x, y \in \mathbb{R}$. So, if we assume (1) for some $a > 0$, then $f(1) = a$ and by rewriting a^x as $(e^{\ln a})^x = e^{(\ln a)x}$, we get

$$f'(x) = e^{(\ln a)x} \ln a = (\ln a)a^x \tag{3}$$

Hence the data $\left. \frac{df(x)}{dx} \right|_{x=1} = 2$ means

$$a \ln a = 2 \tag{4}$$

Since $a = f(1)$, this gives $f(1) \ln f(1) = 2$, which answers the question. (Note that the data does not give the value of a . Nor is it needed to answer the question as it stands.)

There is a flaw in this reasoning. It simply *assumes* that (1) is a solution of (2). The real problem is to show that the only solutions of (2) are of this form.

So now we *derive* that the only solutions of (2) under the hypothesis of the problem are of the form (1). First we show that $f(0) = 1$. Let $a = f(0)$. Then $f(0+0) = f(0)^2$ gives $a^2 = a$. This means either $a = 0$ or $a = 1$. If $f(0) = a = 0$, then for any x we have $f(x) = f(x+0) = f(x)f(0) = 0$ which means that f is identically 0. But then so will be its derivative contradicting that $f'(1) = 2$.

So, $f(0) = 1$. Note that now f never vanishes, for if $f(x) = 0$ (say), then $f(0) = f(x+(-x)) = f(x)f(-x) = 0$ a contradiction. Next we claim that differentiability of f at any one point implies it is differentiable everywhere.

For, consider any $x \in \mathbb{R}$. For any $h \neq 0$, we have $f(x+h) = f(x)f(h)$ by (1). Since $f(0) = 1$, we get

$$\frac{f(x+h) - f(x)}{h} = \frac{f(x)f(h) - f(x)}{h} = f(x) \frac{f(h) - f(0)}{h} \quad (5)$$

Since $f(x) \neq 0$, the limit of the R.H.S. as $h \rightarrow 0$ (with x fixed), exists if and only if the limit of the L.H.S. exists. So, if $f'(0)$ exists then $f'(x)$ exists for all x and if $f'(x)$ exists for some x , then $f'(0)$ exists and hence $f'(y)$ exists for all y .

By letting $h \rightarrow 0$ in (5), we get that $f(x)$ satisfies the differential equation

$$f'(x) = \lambda f(x) \quad (6)$$

where $\lambda = f'(0)$. The value of λ is not given. But it is some constant. This is a first order linear differential equation with constant coefficients. Its general solution is

$$f(x) = ke^{\lambda x} \quad (7)$$

where k is some constant. The initial condition that $f(0) = 1$ gives $k = 1$. (A more direct way to solve (6) is by writing it as $\frac{dy}{dx} = \lambda y$ and then splitting the variables.)

If we now call e^λ as a , then (7) becomes $f(x) = a^x$ for all x . This is precisely (1) which we earlier simply assumed. We are *now* justified in concluding that (B) is correct. ■

Yet another problem absolutely ill suited for an MCQ. Note that continuity of f was not used. However, the functional relation is strong enough to conclude that differentiability of f at even one point implies it is differentiable everywhere.

Functional equations have recently been a fertile source of new problems for post HSC entrance examinations and for various Olympiads. Sometimes such problems tend to be tricky as the equations are weird. The functional equation in the present problem is quite well known and forms the Main Problem of Chapter 20. Unlike differential equations, not many books are available for functional equations. We mention with approval *Functional Equations - A Problem Solving Approach* by B. J. Venkatachala, published by Prism Books Pvt. Ltd.

Q.27 The expression $\sum_{k=0}^{10} 2^k \tan(2^k)$ equals

- (A) $\cot 1 + 2^{11} \cot(2^{11})$
- (B) $\cot 1 - 2^{10} \cot(2^{10})$
- (C) $\cot 1 + 2^{10} \cot(2^{10})$

$$(D) \cot 1 - 2^{11} \cot(2^{11})$$

Answer and Comments: (D). The papersetters have given an implied hint that since the sum comes to only the sum or the difference of two cotangents, the key is probably to express the series on the L.H.S. as a telescopic series so that when you add the terms, most of the summands cancel and only two are left. This was also the gist to the solution of Problem 8 where the key step was to write $\frac{1}{n(n+1)}$ as $\frac{1}{n} - \frac{1}{n+1}$. The difference is that here it is not so obvious how to split the terms.

We can take a cue from the very first term of the summation, viz. $\tan 1$. We would like to express this as $\cot 1$ minus something that would cancel with a part of the next term, viz. $2 \tan 2$. So we first work out the difference $\cot \theta - \tan \theta$. Using standard trigonometric identities for the tangent of twice an angle, we have

$$\begin{aligned} \cot 1 - \tan 1 &= \frac{1}{\tan 1} - \tan 1 \\ &= \frac{1 - \tan^2 1}{\tan 1} \\ &= \frac{1}{\frac{1}{2} \tan 2} \\ &= 2 \cot 2 \end{aligned} \tag{1}$$

More generally, for any θ , we have

$$\cot \theta - \tan \theta = 2 \cot 2\theta \tag{2}$$

Put differently,

$$\tan \theta = \cot \theta - 2 \cot 2\theta \tag{3}$$

This is exactly what we need. Applying this (with changing values of θ) to the terms of the series we get

$$\begin{array}{rcl} \tan 1 & = & \cot 1 - 2 \cot 2 \\ 2 \tan 2 & = & 2 \cot 2 - 4 \cot 4 \\ 4 \tan 4 & = & 4 \cot 4 - 8 \cot 8 \\ & \vdots & \\ 2^r \tan(2^r) & = & 2^r \cot(2^r) - 2^{r+1} \cot(2^{r+1}) \\ & \vdots & \\ 2^{10} \tan(2^{10}) & = & 2^{10} \cot(2^{10}) - 2^{11} \cot(2^{11}) \end{array}$$

When all the terms on the R.H.S. are added, all get canceled except $\cot 1$ and $\cot(2^{11})$. So the given series equals $\cot 1 - 2^{11} \cot(2^{11})$. ■

An easy problem once the idea of recasting the series as a telescopic series using (3) strikes. But there is little point in asking two problems based on telescopic series. Since the telescopic series in Problem 8 is quite well known, that problem could have been dropped.

- Q.28 If the maximum and the minimum values of $\sin^6 x + \cos^6 x$ as x takes all real values are a and b , then $a - b$ equals

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) 1

Answer and Comments: (C). The expression $\sin^6 x + \cos^6 x$ has a factor $\sin^2 x + \cos^2 x$ which is always 1. Dividing by it, the problem amounts to finding the maximum and the minimum on $\mathcal{R}$ of $f(x)$ defined by

$$f(x) = \sin^4 x - \sin^2 x \cos^2 x + \cos^4 x \quad (1)$$

If we now put $u = \sin^2 x$, then u varies over $[0, 1]$ as x moves over $\mathcal{R}$. So we might as well find the extrema of the function

$$g(u) = u^2 - u(1 - u) + (1 - u)^2 = 3u^2 - 3u + 1 \quad (2)$$

This is a simple quadratic with global minimum at $u = \frac{1}{2}$. Hence

$$b = g\left(\frac{1}{2}\right) = \frac{3}{4} - \frac{3}{2} + 1 = \frac{1}{4} \quad (3)$$

For finding the maximum, a , we note that $g(u)$ decreases for $u < \frac{1}{2}$ and increases for $u > \frac{1}{2}$. So the maximum on $[0, 1]$ must occur at one of the end points. Since both $g(0)$ and $g(1)$ equal 1, we get $a = 1$. Therefore $a - b = \frac{3}{4}$. ■

An absolutely simple problem once the idea of factoring $\sin^6 x + \cos^6 x$ and then converting to $g(u)$ strikes. An even easier solution, contributed by Vinayak Antarkar, is to recast $f(x)$ further as $1 - 3\sin^2 x \cos^2 x = 1 - \frac{\sin^2 2x}{4}$. Those who try the problem directly by differentiating $\sin^6 x + \cos^6 x$ will pay heavily in terms of time. Perhaps that was the intention of the papersetters. In terms of degree of difficulty, this problem is the antithesis of Problem 27 of ISI Entrance Test of 2014 asking for the minimum value of $|\sin x + \cos x + \tan x + \operatorname{cosec} x + \sec x + \tan x|$. It turned out that the same question was asked in the 2003 Putnam Competition.

- Q.29 If two real numbers x and y satisfy $(x + 5)^2 + (y - 10)^2 = 196$, then the minimum possible value of $x^2 + 2x + y^2 - 4y$ is

(A) $271 - 112\sqrt{5}$ (B) $14 - 4\sqrt{5}$ (C) $276 - 112\sqrt{5}$ (D) $9 - 4\sqrt{5}$

Answer and Comments: (A). Yet another problem where a function of two variables, viz.

$$f(x, y) = x^2 + 2x + y^2 - 4y \quad (1)$$

is to be minimised subject to the constraint that (x, y) lies on the curve, say C defined by

$$C = \{(x, y) : (x + 5)^2 + (y - 10)^2 = 196\} \quad (2)$$

So, we can adopt the same method that was used in Problem 6, viz. to parametrise C so as to express x and y as functions of a single parameter θ and thereby convert the problem to that of minimising a function of a single variable θ . As C is a circle of radius 14 with centre at $(-5, 10)$ we can parametrise it as

$$x = -5 + 14 \cos \theta, \quad y = 10 + 14 \sin \theta \quad (3)$$

Substitution into (1) gives

$$(-5 + 14 \cos \theta)^2 + 2(-5 + 14 \cos \theta) + (10 + 14 \sin \theta)^2 - 4(10 + 14 \sin \theta)$$

which simplifies to

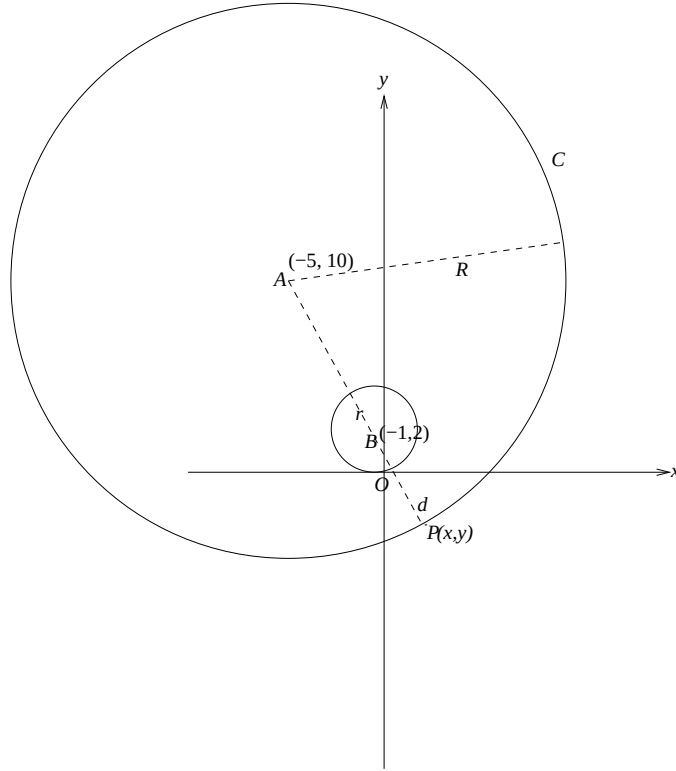
$$271 + 14(-8 \cos \theta + 16 \sin \theta) \quad (4)$$

as the function to be minimised for $\theta \in [0, 2\pi)$. As can be shown by several methods, the minimum value of a linear combination $a \cos \theta + b \sin \theta$ is $-\sqrt{a^2 + b^2}$. So the minimum of $f(x, y)$ is $271 - 14\sqrt{8^2 + 16^2} = 271 - 112\sqrt{5}$. ■

There is an alternate geometric solution too. If we add 5 to $f(x, y)$ we get a new function

$$h(x, y) = f(x, y) + 5 = (x + 1)^2 + (y - 2)^2 \quad (5)$$

Minimising f is as easy as minimising h . But the advantage in working with $h(x, y)$ is that it is the square of the distance of a point, say $P = (x, y)$ from the point $B = (-1, 2)$. So we might as well minimise the distance, say d , of P from B as P varies over the circle C . And this can be done geometrically.



The circle with centre $(-1, 2)$ and radius 2 lies completely inside C . When a circle of radius R contains another circle of radius r inside it, then the point of P on C closest to the inner circle lies on their common diameter at a distance $AB + r$ from A where A and B are the centres. Hence the minimum distance d is given by

$$d = R - (AB + r) \quad (6)$$

In the present problem, $R = 14$, $r = 2$ and $AB = \sqrt{16 + 64} = \sqrt{80}$. So we get $d + r$, i.e. $14 - \sqrt{80}$ as the minimum distance. Therefore its square is the minimum value of $h(x, y)$ as (x, y) moves over C . Since $f(x, y) = h(x, y) - 5$, the minimum value of $f(x, y)$ as (x, y) moves over C is $(14 - \sqrt{80})^2 - 5 = 271 - 112\sqrt{5}$, which is the same answer as before.

The papersetters probably had the second method in their mind as otherwise, in presence of Problem 8, there was little point in asking another problem on constrained optimisation. Even though the geometric solution is easier, the first solution is not prohibitively complicated.

Q.30 Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} (1 - \cos x) \sin(\frac{1}{x}), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then,

- (A) f is discontinuous
- (B) f is continuous but not differentiable
- (C) f is differentiable and its derivative is discontinuous
- (D) f is differentiable and its derivative is continuous

Answer and Comments: (C). The functions f_m defined for non-negative integers m by

$$f_m(x) = \begin{cases} x^m \sin(\frac{1}{x}), & x \neq 0 \\ 0, & x = 0 \end{cases} \quad (1)$$

are standard examples where the smoothness of f_m increases with m . All these functions have derivatives of all orders at all non-zero values of x . But at 0, f_0 is not even continuous, f_1 is continuous but not differentiable, f_2 is differentiable but the derivative f'_2 not continuous, f_3 is differentiable and the derivative f'_3 is continuous but not differentiable at 0, f_4 as well as f'_4 are differentiable and so on. The proofs are technical. But the essential idea is that the power x^m 'cures' the irregularity of $\sin(\frac{1}{x})$ at 0 and the higher the integer m , the stronger is this cure.

In the present problem, instead of x^m we have $1 - \cos x$ as the coefficient of $\sin(\frac{1}{x})$. Near 0, $1 - \cos x$ is comparable to x^2 in the sense that the ratio $\frac{1 - \cos x}{x^2}$ tends to a finite non-zero limit as $x \rightarrow 0$. (A proof of this can be given using that $1 - \cos x = 2 \sin^2(\frac{x}{2})$ and the fact that $\frac{\sin \theta}{\theta}$ tends to 1 as $\theta \rightarrow 0$. It is tempting to get an easier proof using l'Hôpital's rule. But the ease gained is illusory because this rule requires the derivatives of trigonometric functions and these derivatives are obtained by using that $\frac{\sin \theta}{\theta} \rightarrow 1$ as $\theta \rightarrow 0$.)

Once the comparability of $1 - \cos x$ with x^2 is proved, the given function will behave exactly like the function f_2 above. As already noted, f_2 is differentiable at 0 but the derivative is not continuous at 0. So this is also true of the given function $f(x)$. ■

A simple problem once the idea of comparability of $1 - \cos x$ with x^2 strikes.

ISI BStat-BMath-UGA-2021 Paper 2 with Comments

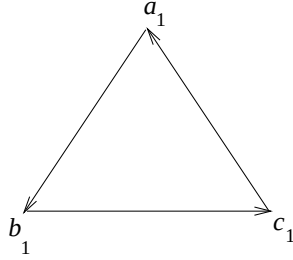
Q.1 There are three cities each of which has the same number of citizens, say n . Every citizen in each city has exactly a total of $n + 1$ friends in the other two cities. Show that there exist three people, one from each city, such that they are friends. We assume that friendship is mutual (that is, a symmetric relation).

Solution and Comments: Call the three cities as A, B, C and denote their citizens by respective small case letters, with subscripts if necessary. We do not count how many friends a citizen has in his own city as this is not relevant to the conclusion.

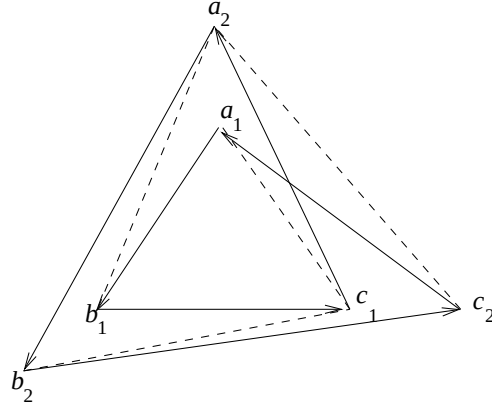
As the total number of friends a person has is more than the number of citizens in any city, it follows that every $a \in A$ has at least one friend in each of B and C . The same is true for citizens of B and C .

It is tempting to try induction on n . The case $n = 1$ is trivially true as there are only 3 persons and every two of them are friendly by hypothesis. But there will be problem in the inductive step. The most natural attempt would be to remove one citizen from each city. Suppose these persons are a, b, c . But after their removal, a person who is friendly with two of these will lose two friends and so his number of friends will now be $n + 1 - 2 = n - 1$, whereas for the inductive hypothesis to apply it will have to be n .

Another approach is to start with any person and go adding friends till you get back to the person you start with. It is convenient to express this in terms of graphs. Consider a graph G with $3n$ vertices corresponding to the citizens. Join two friends by an edge. Now suppose we start with some a_1 and take any adjacent b_1 and then any c_1 adjacent to b_1 . If luckily a_1 and c_1 are adjacent then we get a cycle of length 3 as shown in (a) of the figure on the next page. This is a desired cycle. (The edges are shown by straight lines although they could have been any arcs. Also since friendship is symmetric there is no need to put an arrow on the edges. They are indicative of the sequence in which we approach the vertices. Absence of friendship is shown by a dotted line as in (b).)



(a)



(b)

When c_1 is not adjacent to a_1 , let a_2 be any vertex in A adjacent to c_1 . (It exists because c_1 is adjacent to at least one a_i , for $i \neq 1$.) If a_2 is adjacent to b_1 , we get a 3-cycle $a_2 - b_1 - c_1 - a_2$. If not, join a_2 to some b_2 . Again if b_2 is adjacent to c_1 we are done. Otherwise join b_2 to some c_2 . Once again, if c_2 is adjacent to a_2 , then $a_2 - b_2 - c_2 - a_2$ is a desired 3-cycle. Otherwise c_2 is adjacent to some a_i with $a_i \neq 2$. If $i > 2$ we relabel a_i as a_3 and continue. But it may happen that $i = 1$. There is nothing in the problem to preclude this. In that case we get a cycle $a_1 - b_1 - c_1 - a_2 - b_2 - c_2 - a_1$ as shown in (b). But this is a cycle of length 6 and not 3. Nor is there any way to split it into two cycles of length 3.

Here we started with an arbitrary vertex. It turns out that if the starting vertex is chosen carefully, then a cycle of length 3 can be found. Here we choose it by applying what is called the **extremality principle**. (See the comments on the solution to Problem 5 in Paper 2 of ISI 2019.) That is, we choose the starting vertex which is maximal (or minimal) in some sense. In the graph constructed above, the degree of every vertex is given to be the same, viz. $n + 1$. In the original setting $n + 1$ is the total number of friends every person has. But their distribution may vary between the two cities. If $a \in A$ has d_1 vertices adjacent in B and d_2 in C , we call $\max\{d_1, d_2\}$ the **major degree** of a and $\min\{d_1, d_2\}$ as its **minor degree**. (Note that $d_1 + d_2$ is always $n + 1$ and when n is odd it may happen for some vertex that its major and minor degrees are equal each being $\frac{n+1}{2}$. However, in general, the minor degree cannot exceed the major degree. For a vertex in B or C the major and minor degrees are defined analogously. (In terms of the original setting of the problem, the major degree of a person is the largest number of friends he has in a single city.)

As the number of vertices is finite, among all vertices there is some whose major degree is largest (and hence also whose minor degree is smallest).

Without loss of generality, assume this vertex is some $a \in A$. Let r be its major degree (so that $r \geq \frac{n+1}{2}$) and without loss of generality assume that the r single city friends of a are in C , say $c_1, c_2, \dots, c_r$. Now let b be any vertex in B adjacent to a . If b is adjacent to c_i for some $i \leq r$, then $a - b - c_i - a$ is a desired cycle of length 3. If b is not adjacent to any c_i , then the number of friends b has in C is at most $n - r$. But that means that the minor degree of b is at most $n - r$. Since the minor degree of a is $n + 1 - r$, we get that b has a smaller minor degree than a , contradicting the choice of a . This contradiction proves the assertion. ■

The maximality principle sometimes provides the right starting point when none other is obvious. See the solution to the JEE 1982 problem in Comment No. 15 of Chapter 13.

Q.2 Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be a function satisfying $f(0) \neq 0 = f(1)$. Assume also that f satisfies conditions (A) and (B) below.

$$(A) \quad f(xy) = f(x) + f(y) - f(x)f(y)$$

$$(B) \quad f(x - y)f(x)f(y) = f(0)f(x)f(y)$$

- (i) Determine explicitly the set $f(a) : a \in \mathbb{Z}$.
- (ii) Assuming that there is a non-zero integer a , such that $f(a) \neq 0$, prove that the set $b : f(b) \neq 0$ is infinite.

Solution and Comments: Like Problem 26 in Paper 1, this is a problem on functional equations. There is also some similarity in the form of the functional equation. In Problem 26, values of $f(x + y)$ were related to those of $f(x)$ and $f(y)$. Here the values of $f(xy)$ are related to those of $f(x)$ and $f(y)$. But in Problem 26, solutions of the form $f(x) = a^x$ were obvious. In this problem there is no such obvious solution. Nor is it asked. But in Problem 26, without assuming any form of the solution, we began by giving special values to the variables. Here we try something similar.

Let us call $f(0)$ as c . If we put $x = 0 = y$ in (A), we get an equation for c , viz.

$$c = 2c - c^2 \tag{1}$$

which has exactly two solutions, $c = 0$ and $c = 1$. The first is precluded by hypothesis. So $c = f(0) = 1$. (This was also the first step in deriving the solution to the equation in Problem 16, except that there the possibility $f(0) = 0$ was ruled out as leading to the identically zero function.)

Since $f(0) = 1$, putting $y = 0$ in (B) gives

$$f(x)f(x) = f(x) \tag{2}$$

for all $x \in \mathbb{Z}$. So, once again the only possible values of $f(x)$ are 0 and 1. Since it is given that $f(1) = 0$ and we have shown $f(0) = 1$, the range of f is only the set $\{0, 1\}$. This answers (i).

For (ii), if $f(a) \neq 0$, then $f(a) = 1$ since the only possible values assumed by f are 0 and 1. Putting $x = y = a$ in (A), we have

$$f(a^2) = 1 + 1 - 1 = 1 \quad (3)$$

So starting with a , we get a^2 as another solution of $f(x) = 1$. Since $a \neq 0$ and $a \neq 1$ by hypothesis a^2 is a different solution than a . Starting from a^2 , we get a^4 as yet another solution. Continuing $a^4, a^8, a^{16}, \dots$ all satisfy $f(x) = 1$. Further, since $|a| > 1$, these numbers are all distinct as their absolute values are strictly increasing. So there are infinitely many values of b for which $f(b) = 1$. ■

The point of asking two problems on functional equations is not clear. It is interesting that in the solutions to both, we used the fact that the only solutions of $y^2 = y$ are 0 and 1. An element which equals its own square (and hence also all its higher powers) is called an **idempotent**. In numbers, 0 and 1 are the only idempotents. But the concept makes sense whenever there is an operation of multiplication. For example, a square matrix A is called idempotent if $A^2 = A$. For $n > 1$, there are many idempotent matrices of order n other than the zero matrix and the identity matrix.

Q.3 Prove that every positive rational number can be expressed uniquely as a finite sum of the form

$$a_1 + \frac{a_2}{2!} + \frac{a_3}{3!} + \dots + \frac{a_n}{n!}$$

where a_n are integers such that $0 \leq a_n \leq n - 1$ for all $n > 1$.

Solution and Comments: A minor notational discrepancy is that since n is the index of the last term, the restriction should have been given as a_k are integers with $0 \leq a_k \leq k - 1$ for all $k = 2, 3, \dots, n$.

Let x be a positive rational number and assume that it can be expressed as asked i.e. as

$$x = a_1 + \frac{a_2}{2!} + \frac{a_3}{3!} + \dots + \frac{a_n}{n!} \quad (1)$$

where a_k are integers such that $0 \leq a_k \leq k - 1$ for all $k > 1$. Note that there is no restriction on a_1 . It could be any integer. But there is a restriction that $0 \leq a_k \leq k - 1$ for all $k > 1$. As a result, if we denote by S the sum of the remaining terms in the R.H.S., we get an upper bound on it, viz.

$$\begin{aligned} S &= \frac{a_2}{2!} + \frac{a_3}{3!} + \dots + \frac{a_n}{n!} \\ &\leq \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{n-1}{n!} \end{aligned} \quad (2)$$

This can be recognised as a telescopic series if we write $\frac{k-1}{k!}$ as $\frac{1}{(k-1)!} - \frac{1}{k!}$ for $k = 2, 3, \dots, n$. Therefore

$$S \leq 1 - \frac{1}{n!} \quad (3)$$

Hence S lies in $[0, 1)$. Therefore given a positive rational x , we can take a_1 as its integral part $[x]$ and the problem now reduces to showing that the fractional part $\{x\}$ can be expressed in the form given.

The situation is analogous to expressing a real number $y \in [0, 1)$ in its decimal expansion, that is

$$y = \sum_{k=1}^{\infty} \frac{y_k}{10^k} \quad (4)$$

where y_k are integers satisfying

$$0 \leq y_k \leq 9 \quad (5)$$

for $k = 1, 2, \dots$. Here the sum can be infinite. (If y is rational then it can be shown that the series is finite if and only if the denominator of y (in its reduced form) has no prime factors other than 2 and 5. But that is not vital here.)

The process of obtaining this expansion can be simulated by a physical example. Put a piece of mass y in pan A of a balance. We have to put a suitable pile of pieces of standard masses in the other pan B till the two pans balance each other. The pieces are of masses $\frac{1}{10}, \frac{1}{100}, \frac{1}{1000}, \dots$ and we can take at most 9 pieces of each type.

To begin with, we use pieces of mass $\frac{1}{10}$. If luckily r of these balance y exactly then $y = \frac{r}{10}$ and its decimal expansion is $0.r$. Otherwise we let r_1 be the largest integer (from 0 to 9) such that $y > \frac{r_1}{10}$. Then we replace y by $y - \frac{r_1}{10}$ and try to balance it with pieces of mass $\frac{1}{100}$ each. Once again let r_2 be the largest integer (from 0 to 9) such that $y - \frac{r_1}{10} \geq \frac{r_2}{100}$. If equality holds, the expansion of y is $0.r_1 r_2$. Otherwise we balance $y - \frac{r_1}{10} - \frac{r_2}{100}$ with pieces of mass $\frac{1}{1000}$ each. This process continues.

In principle, our task is similar except that the standard masses we have are of the form $\frac{1}{2}, \frac{1}{6}, \dots, \frac{1}{k!}, \dots$ and the restriction is that you can take at most $k - 1$ pieces of mass $\frac{1}{k!}$. For example, let $x = \frac{7}{10}$. Since $x > \frac{1}{2!}$ we select one piece of mass $\frac{1}{2!}$ to go into pan B . Now we have to balance $\frac{7}{10} - \frac{1}{2} = \frac{1}{5}$ using smaller masses. Since $\frac{1}{6} < \frac{1}{5} < \frac{2}{6}$, we select one piece of mass $\frac{1}{6}$ to go into pan B . Now have to make up $\frac{1}{5} - \frac{1}{6} = \frac{1}{30}$. The next available mass $\frac{1}{24}$ is already bigger than $\frac{1}{30}$. So we take no pieces of this type. We are still left with $\frac{1}{30}$ and the next mass type available is $\frac{1}{5!} = \frac{1}{120}$. Luckily, 4 such pieces make a total of exactly $\frac{1}{30}$. So we have

$$\frac{7}{10} = \frac{1}{2} + \frac{1}{5} = \frac{1}{2!} + \frac{1}{6} + \frac{1}{30} = \frac{1}{2!} + \frac{1}{3!} + \frac{0}{4!} + \frac{4}{5!} \quad (6)$$

as the desired expansion of $\frac{7}{10}$ using multiples of $\frac{1}{k!}$, $k = 2, 3, 4, \dots$ with the restrictions given.

The trouble is to ensure that this process terminates. In the case of a decimal expansion of a real number $x \in [0, 1)$, it will go on forever when x is irrational. Even if x is rational, it may not terminate as we see from $\frac{1}{3} = 0.333333\dots$. It is remarkable that the expansion asked in the problem always terminates in a finite number of steps when x is rational. But if we express x in the manner above, i.e. by subtracting suitable multiples of $\frac{1}{2!}, \frac{1}{3!}, \frac{1}{4!}, \dots$, it is not easy to show that it will terminate.

But we can make full use of rationality of x if we go in the other direction, i.e. from pieces of smaller masses to the larger ones. As a real life analogy, if somebody pays the exact price x of some item but with lots of small coins, we convert as many of these coins as possible to one rupee notes. Even the number of such notes may be too large to carry. Then we exchange 10 of them at a time to notes of Rs.10 till we are left with less than 10 one rupee notes. If necessary, we can exchange tens of such 10 rupee notes for 100 rupee notes and so on.

So, suppose $x = \frac{p}{q}$ is a positive rational in $[0, 1)$ in its reduced form. Then p, q are positive integers with $p < q$. Since $x = \frac{p}{q} = \frac{p(q-1)!}{q!}$, x is the total mass of a pile of $p(q-1)!$ pieces of mass $\frac{1}{q!}$ each. We retain as few of them as permissible and convert the rest to pieces of mass $\frac{1}{(q-1)!}$ each. For this we use the division algorithm (mentioned in the solution to Problem 13 of Paper 1). That is, we divide the integer $p(q-1)!$ by q to get

$$p(q-1)! = bq + r \quad (7)$$

where b and r are integers and $0 \leq r < q$. We now retain r pieces of mass $\frac{1}{q!}$ each and convert the bq pieces into pieces of higher mass, viz. $\frac{1}{(q-1)!}$. In a mathematical form,

$$\begin{aligned} x = \frac{p}{q} &= \frac{p(q-1)!}{q!} \\ &= \frac{bq + r}{q!} \\ &= \frac{b}{(q-1)!} + \frac{r}{q!} \end{aligned} \quad (8)$$

So we now have only r pieces of mass $\frac{1}{q!}$ and b pieces of mass $\frac{1}{(q-1)!}$ each. r is within the permissible range. But b may be a huge number. We are allowed to retain at most $q-2$ pieces of size $\frac{1}{(q-1)!}$. So, this time we reduce b modulo $q-1$ and write

$$b = c(q-1) + s \quad (9)$$

where c, s are integers and the remainder s lies in $\{0, 1, \dots, q-2\}$.

So now we have r pieces of mass $\frac{1}{q!}$, s pieces of mass $\frac{1}{(q-1)!}$ each and c pieces of mass $\frac{1}{(q-2)!}$ each. If that is too many, that is, if $c > q-2$, we reduce c modulo $q-2$. Continuing this process, which is sure to terminate as the denominators are strictly descending, we get the expansion of x of the desired type. Note that the first remainder r is precisely a_q , the second remainder is a_{q-1} and so on. So we are determining the coefficients in the expansion (1) in the reverse order for $k > 1$.

Let us verify that we get the same expansion of $\frac{7}{10}$ as (6) by this method. Here $p = 7$ and $q = 10$. So we determine a_{10} first. It is the remainder left when $7 \times 9!$ is divided by 10. Since 10 divides $9!$, $a_{10} = 0$. The quotient is $7 \times 3 \times 4 \times 6 \times 7 \times 8 \times 9$. This is also a multiple of the next divisor, viz. 9. So a_9 is also 0 and the quotient is $7 \times 3 \times 4 \times 6 \times 7 \times 8$. This is also a multiple of the next divisor 8 and so $a_8 = 0$. This will go on until we are left with 84 pieces of mass $\frac{1}{5!}$. Now 5 does not divide 84. We write $84 = 16 \times 5 + 4$. So $a_4 = 4$ and we are left with 16 pieces of mass $\frac{1}{4!}$ each. This time 16 is a multiple of 4 and so $a_3 = 0$ and we are left with 4 pieces of mass $\frac{1}{3!}$. Going one more step further we get the same expansion as (6).

Now we have proved the existence of a representation of a rational number. The proof of uniqueness is the same as that for decimal expansions. The basic idea is that even if all permissible smaller masses are put together, they cannot match a single large mass. Thus in the case of a decimal expansion, even if we add $\sum_{r=n+1}^{\infty} \frac{9}{10^r}$ we barely get $\frac{1}{10^n}$. Even if one term is less, it falls short. (That is why, sometimes trailing 9's are not allowed in decimal expansions.)

In our case, the situation is similar. Suppose x has two expansions, say

$$x = a_1 + \frac{a_2}{2!} + \frac{a_3}{3!} + \dots + \frac{a_n}{n!} \quad (10)$$

$$\text{and } x = b_1 + \frac{b_2}{2!} + \frac{b_3}{3!} + \dots + \frac{b_n}{n!} \quad (11)$$

are two different expansions. Let r be the smallest integer for which $a_r \neq b_r$ and without loss of generality assume that $a_r > b_r$. Then the difference in the r -th terms, viz. $\frac{a_r - b_r}{r!}$ is at least $\frac{1}{r!}$ and cannot be made up by the remaining terms because the maximum the differences between the remaining terms can add is $\frac{r}{(r+1)!} + \frac{r+1}{(r+2)!} + \dots + \frac{n-1}{n!}$ and by the telescopic nature of the series this sum is $\frac{1}{r!} - \frac{1}{n!}$. (The argument is similar to the derivation of (3).) ■

An excellent problem. The analogy with decimal expansions is helpful but not adequate. The problem is of the same spirit as the **factorial representation** of positive integers given in Exercise (1.11). But it does not appear that a solution can be based directly on it. What is crucially

needed in the solution is the euclidean algorithm (which is also needed in the factorial representation).

- Q.4 Let $g : (0, \infty) \rightarrow (0, \infty)$ be a differentiable function whose derivative is continuous and such that $g(g(x)) = x$ for all $x > 0$. If g is not the identity function, prove that g must be strictly decreasing.

Solution and Comments: The condition

$$g(g(x)) = x \quad (1)$$

means that g is a bijection which is its own inverse. In particular, it is one-to-one. By the Intermediate Value Property (IVP) any injective continuous function (over an interval) is either strictly increasing or strictly decreasing. (See Comment No. 6 of Chapter 16 for a proof.) The problem now amounts to showing that if g is strictly monotonically increasing, then it must be the identity function, i.e. $g(x) = x$ for all x . Suppose not. Then $g(x) \neq x$ for some x . Call $g(x)$ as y . Either $y > x$ or $y < x$. If $y > x$, then $g(y) > g(x)$ by strict monotonicity of g . But that means $g(g(x)) > g(x)$, i.e. $x > g(x)$, a contradiction. A similar contradiction arises if we assume $y < x$. ■

The hypothesis that g is continuously differentiable is nowhere needed. In fact, g need not even be differentiable. The condition $g(g(x)) = x$ is very strong. Although the identity function is the only example where g is strictly increasing, there are many examples besides the function $g(x) = \frac{1}{x}$ when it is strictly decreasing. For example, $g(x) = 1 - \ln x$ for $x \in (0, 1]$ and $g(x) = e^{1-x}$ for $x \in [1, \infty)$. (Contributed by Siddhesh Naik.)

- Q.5 Let $a_0, a_1, \dots, a_{19} \in \mathbb{R}$ and

$$P(x) = x^{20} + \sum_{i=0}^{19} a_i x^i \quad x \in \mathbb{R}$$

If $P(x) = P(-x)$ for all $x \in \mathbb{R}$ and $P(k) = k^2$, for all $k = 0, 1, 2, \dots, 9$, then find

$$\lim_{x \rightarrow 0} \frac{P(x)}{\sin^2 x}$$

Solution and Comments: $1 - (9!)^2$. This is a combination of several standard tricks. First, the trigonometric function $\sin x$ has no role other than that it is comparable to x for x near 0, so that $\lim_{x \rightarrow 0} \frac{P(x)}{\sin^2 x}$ is the same as $\lim_{x \rightarrow 0} \frac{P(x)}{x^2}$. (For a formal proof, write $\frac{P(x)}{\sin^2 x}$ as $\frac{P(x)}{x^2} \times \frac{x^2}{\sin^2 x}$ and observe that the second factor tends to 1 as $x \rightarrow 0$.)

We are given that $P(x)$ is a monic polynomial of degree 20. To determine it uniquely we need 20 equations in its coefficients $a_0, a_1, \dots, a_{19}$. Here the condition $P(x) = P(-x)$ for all $x \in \mathbb{R}$ means that $P(x)$ is an odd function and so all odd degree coefficients are 0. That is, $a_i = 0$ for $i = 1, 3, 5, \dots, 17, 19$. This gives 10 equations. $P(0) = 0$ gives

$$a_0 = 0 \quad (2)$$

The remaining 9 equations come from $P(k) = k^2$ for $k = 1, 2, \dots, 9$. In full form, this is a system of 9 equations:

$$\begin{aligned} a_2 + a_4 + \dots + a_{18} + 1 &= 1 \\ 4a_2 + 16a_4 + \dots + 2^k a_{2k} + \dots + 2^{18} a_{18} + 2^{20} &= 4 \\ &\vdots \\ 9^2 a_2 + 9^4 a_4 + \dots + 9^{18} a_{18} + 9^{20} &= 81 \end{aligned}$$

Our interest is only in a_2 because since $a_0 = a_1 = 0$, $\frac{P(x)}{x^2} = a_2 + a_4 x^2 + \dots + a_{18} x^{16} + x^{18}$ which tends to a_2 as $x \rightarrow 0$ since all other terms tend to 0.

But even though our interest is only in a_2 , getting it by solving the system above is fairly complicated. (It can be shown that the system has a unique solution. But a formula for a_i is complicated.)

Here we adopt a trick which is by now common in JEE type tests. We consider a new polynomial $Q(x)$ defined by

$$Q(x) = P(x) - x^2 \quad (3)$$

We already know that 0 is a root of this polynomial. Further for every $k = 1, 2, \dots, 9$, $\pm k$ are roots because $P(\pm k) = k^2$. Hence $Q(x)$ has x and $(x \pm k)$ as factors. They together multiply to $x(x^2 - 1)(x^2 - 4) \dots (x^2 - 81)$. Since $P(x)$ is monic and of degree 20, so is $Q(x)$. Therefore $Q(x)$ has one more factor which is of the form $x + c$ for some c . We thus get

$$\begin{aligned} P(x) &= x^2(x^2 - 1)(x^2 - 4) \dots (x^2 - 81) \\ &\quad + cx(x^2 - 1)(x^2 - 4) \dots (x^2 - 81) + x^2 \end{aligned} \quad (4)$$

The second product consists of all odd degree terms in $P(x)$. As these are all 0, we drop them. Dividing by x^2 then gives

$$\frac{P(x)}{x^2} = 1 + (x^2 - 1)(x^2 - 4) \dots (x^2 - 81) \quad (5)$$

Hence

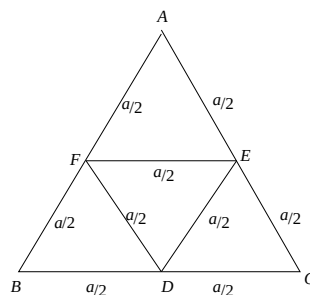
$$\begin{aligned} \lim_{x \rightarrow 0} \frac{P(x)}{x^2} &= 1 + (-1)(-4) \dots (-81) \\ &= 1 - (9!)^2 \end{aligned} \quad (6)$$

As commented earlier, this is also the answer to the problem. ■

There is a general result called **Lagrange's interpolation** which says that given any n distinct real numbers $x_1, x_2, \dots, x_n$ and any (not necessarily distinct) real numbers $y_1, y_2, \dots, y_n$, there exists a unique polynomial $p(x)$ of degree $n-1$ (or less) such that $p(x_i) = y_i$ for $i = 1, 2, \dots, n$. The present problem is based on this. But the general construction of the polynomial is fairly complicated. If some of the y_i 's are 0, the work is simplified because then we know some factors of $p(x)$. Sometimes the data is so adjusted that by introducing some auxiliary polynomial $f(x)$, the difference $p(x) - f(x)$ is a polynomial which is much easier to find. This trick was used in designing Problem 1 of Paper 1 of ISI B. Stat. Entrance 2019. (See the slick alternate solution given at the end in the comments.) The present problem is of this type where the auxiliary polynomial is x^2 .

- Q.6 If an equilateral triangle $\triangle$ of side length a lies in the union of five equilateral triangles of side length b , show that there exist four equilateral triangles of side length b whose union contains $\triangle$.

Solution and Comments: Let ABC be the triangle $\triangle$ and D, E, F the midpoints of its sides as shown in the accompanying diagram. These points and the lines joining them divide $\triangle$ into four equilateral triangles of side $a/2$ each.



We are given that T is contained in the union of five equilateral triangles, say T_1, T_2, T_3, T_4 and T_5 , each with side b . By the pigeon hole principle at least two of these six points are in one of these, say in T_1 . Since T_1 is a convex set, the entire segment joining these two points is contained in T_1 . But no matter which these points are, the distance between them is at least $a/2$. So, the equilateral triangle T_1 contains a line segment of length $a/2$. But the longest line segment contained in T_1 is any of its side. Therefore we get $b \geq a/2$. By construction, $\triangle$ is the union of four triangles of side $a/2$ each. By enlarging these four triangles if necessary, we get four equilateral triangles whose union contains $\triangle$. (Note that the question does not ask to show that $\triangle$ is contained in the union of four of the original equilateral triangles of side b .) ■

A simple problem based on the pigeon hole principle and the convexity of every triangle.

- Q.7 Let a, b, c be three real numbers which are roots of a cubic polynomial and

satisfy $a + b + c = 6$ and $ab + bc + ca = 9$. Suppose $a < b < c$. Show that $0 < a < 1 < b < 3 < c < 4$.

Solution and Comments: As it stands, the problem is about inequalities. Specifically, three real numbers a, b, c are given to satisfy the equations

$$a + b + c = 6 \quad (1)$$

$$\text{and } ab + bc + ca = 9 \quad (2)$$

and the inequalities

$$a < b < c \quad (3)$$

and we are asked to prove that

$$0 < a < 1 < b < 3 < c < 4 \quad (4)$$

There is no need to mention that a, b, c are the roots of a cubic polynomial. This way, the papersetters have given an implicit hint that properties of cubic functions (including those where powerful methods from calculus are used) may be applicable. Even without this hint, the very form of (1) and (2) would suggest a perceptive reader to consider a cubic with roots a, b, c . But this test of perceptivity has been obviated.

Let us first see how far we can go without involving any cubic polynomial. Note that it is not given that a, b, c are non-negative. Hence standard tools like the A.M.-G.M. inequality cannot be used right away. In general two equations in three unknowns do not determine them (except in accidental situations such as Problem 15 in Paper 1). Nevertheless, any two of a, b, c give only two possible values of the third. To see this we get $b + c = 6 - a$ from (1). Putting this into (2) and simplifying, we have

$$(a - 3)^2 = bc \quad (5)$$

and by a similar reasoning

$$(b - 3)^2 = ca \quad (6)$$

$$\text{and } (c - 3)^2 = ab \quad (7)$$

We know now that every two of a, b, c are of the same sign which is non-negative by (1). The possibility of any one, say a , vanishing gives $b = c = 3$ from (5) and (6) respectively. These are degenerate solutions and do not satisfy (3) (which, in particular implies that a, b, c are distinct).

Thus now we know that a, b, c are all positive. The real task is to prove that they lie in the open intervals $(0, 1)$, $(1, 3)$ and $(3, 4)$ respectively. For this we introduce a cubic $f(x)$ with roots a, b, c . We do not know abc . Still, from (1) and (2), we can take

$$f(x) = x^3 - 6x^2 + 9x - abc \quad (8)$$

By Rolle's theorem, the intervals (a, b) and (b, c) each contain at least one zero of f' . But

$$f'(x) = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3) \quad (9)$$

Since we are given $a < b < c$, we get that $1 \in (a, b)$ and $3 \in (b, c)$. As we have already proved that a, b, c are all positive, we now have

$$0 < a < 1 < b < 3 < c \quad (10)$$

which is almost the same as (4) except that we still have to prove that $c < 4$. This is where we need a deeper understanding of the critical points of a cubic. As the leading coefficient of $f(x)$ is positive, the smaller critical point 1 is a local maximum while the larger one, viz. 3, is a point of local minimum. From (9), we have that f is strictly increasing on $(-\infty, 1]$, strictly decreasing on $[1, 3]$ and again strictly increasing on $[3, \infty)$. As both c and 4 are in this last interval, showing that $c < 4$ is equivalent to showing that $f(c) < f(4)$. $f(c)$ is given to be 0. By a direct calculation,

$$f(4) = 64 - 96 + 36 - abc = 4 - abc \quad (11)$$

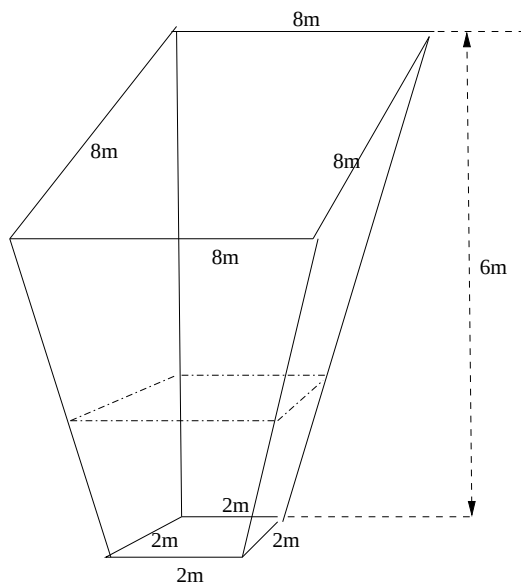
So we shall be through if we can prove $4 - abc > 0$, or equivalently, that $abc < 4$. This does not follow directly from the bounds we have found on a and b . Choosing a and b subject to these bounds we can make $ab > 1$ and hence $c = 3 + \sqrt{ab} > 4$.

However, we observe that $f(1)$ also equals $4 - abc$. Since we already know that $a < 1$ and f is strictly increasing on $(-\infty, 1]$, we have $f(a) < f(1)$, i.e. $4 - abc > 0$ as was to be proved. So $c < 4$ also holds. ■

It may appear a pleasant coincidence that $f(1) = f(4)$. But that is probably how the numerical data was drafted, as otherwise there is no particular role for the number 4.

There is an alternate way to prove that $abc < 4$, using the same reasoning as in the solutions to Q.16 or Q.18 of Paper 1. a, b, c are the distinct roots of the cubic equation $g(x) = \lambda$ where $g(x) = x^3 - 6x^2 + 9x$ and $\lambda = abc$. The function $g(x)$ has a local minimum at $x = 3$ and a local maximum at $x = 1$. Since $g(3) = 0$ and $g(1) = 4$, it is only for $0 < \lambda < 4$, that three distinct solutions exist. Hence, in particular, $\lambda < 4$, i.e. $abc < 4$.

- Q.8 A pond has been dug at the Indian Statistical Institute as an inverted truncated pyramid with a square base (see figure below). The depth of the pond is 6 m. The square at the bottom has side length 2 m and the top square has a side length of 8 m. Water is filled in at a rate of $\frac{19}{3}$ cubic meters per hour. At what rate is the water level rising exactly 1 hour after the water started to fill the pond ?



Solution and Comments: $\frac{19}{27}$ m./hr.

A typical problem on related rates. Let t denote time (in hours) measured from the moment the filling started. We are given that

$$\frac{dV}{dt} = \frac{19}{3} \quad (1)$$

where $V = V(t)$ is the volume (in cubic meters) in the pond at time t . Let $h = h(t)$ denote the height (in meters) of the water at time t . We are asked the value of $\frac{dh}{dt}$ at $t = 1$. If we can express V in terms of h , then we can use the chain rule and apply (1).

When the height of the cross section of the pond changes from 2 m. to 8 m., the side of its cross section also changes from 2 m. to 8 m. (One such cross section is shown in the figure with dotted boundary.) So the cross section at a height h from the base is a square of side $h + 2$ m. V is the volume below it and can be thought of as the difference between two coaxial pyramids, one with height and side $h + 2$ m. and the other with height and side 2 m. So,

$$V = \frac{1}{3}[(h + 2)^3 - 8] = \frac{1}{3}(h^3 + 6h^2 + 12h) \quad (2)$$

(Actually, we can get $\frac{dV}{dh}$ more easily without expanding.)

$$\frac{dV}{dh} = h^2 + 4h + 4 = (h + 2)^2 \quad (3)$$

By the chain rule

$$\frac{dV}{dt} = \frac{dV}{dh} \frac{dh}{dt} \quad (4)$$

$$= (h+2)^2 \frac{dh}{dt} \quad (5)$$

The L.H.S. is given to be a constant $\frac{19}{3}$. So we get a differential equation for $h = h(t)$, viz.

$$(h+2)^2 \frac{dh}{dt} = \frac{19}{3} \quad (6)$$

with the initial condition

$$h(0) = 0 \quad (7)$$

Splitting the differentials,

$$(h+2)^2 dh = \frac{19}{3} dt \quad (8)$$

Integrating both the sides

$$\frac{(h+2)^3}{3} = \frac{19}{3}t + c \quad (9)$$

where c is some constant. The initial condition gives

$$\frac{8}{3} = c \quad (10)$$

Hence

$$(h+2)^3 = 19t + 8 \quad (11)$$

As we want $\frac{dh}{dt}$, we express h directly as a function of t .

$$h = (19t + 8)^{\frac{1}{3}} - 2 \quad (12)$$

Therefore

$$\frac{dh}{dt} = \frac{19}{3}(19t + 8)^{-2/3} \quad (13)$$

At $t = 1$, the R.H.S. is $\frac{19}{3} \frac{1}{(27)^{2/3}} = \frac{19}{3} \frac{1}{9} = \frac{19}{27}$.

So the water level is rising at the rate of $\frac{19}{27}$ m./hr. ■

A similar problem (but with draining rather than filling a tank which was hemispherical and more complicated because the rate of drainage was not constant) was asked in JEE 2001. (See the Main Problem of Chapter 19.) In those days full length questions were allowed. Such a problem can be artificially converted to an MCQ. But the work demanded is far too much and prone to slips of calculation. So such problems went out of focus after JEE became totally MCQ type. It is good that ISI Entrance Test has accommodated a problem of this type through its second paper.

CONCLUDING REMARKS

The number of good problems in both the papers has improved. Especially novel problems from Paper 1 are :

1. Problem 3: Combination of an infinite series and the integral of a not quite periodic function.
2. Problem 7: Monotonicity of the function $\sin x + \alpha x$.
3. Problem 11: Limit of a function defined by an integral.
4. Problem 13: g.c.d. of two polynomials in a prime
5. Problem 14: Sum of the determinants of all matrices in a certain class.
6. Problem 15: Sum of the squares of four non-negative numbers which add to 1.
7. Problem 19: Superficially, a problem about Cauchy-Euler type differential equations.
8. Problem 20: Images of circles under complex reciprocation.
9. Problem 22: A function of the order of x^2 near 0.
10. Problem 25: Solutions of a trigonometric equation.

Out of these, Problems 15 and 19 are tricky. Problem 15 makes a novel use of inequalities in a cyclic order. Although there are several problems on number theory, except for Problem 13 in Paper 1, they do not measure up to the mark. Some of them need only parity consideration. Some of these could have been dropped and replaced by problems in combinatorics or problems of evaluating some interesting limits or problems based on comparative rates of growth. Combinatorics and probability have been paid only a lip service.

There is also considerable duplication of ideas. For example, telescopic series appear in as many as two problems in Paper 1 and one in Paper 2. Constrained optimisation has also appeared in two problems. Functional equations have appeared in two. Both Problems 16 and 18 are based on the same idea, viz. interpreting solutions of an equation of the form $f(x) = c$ in terms of the points of intersection of a horizontal line with the graph of $f(x)$. (Moreover, this idea can also be used to show that $abc < 4$ in Q.7 of Paper 2.) Problem 19 is an excellent problem based on convexity of a function. There was little point in asking Problem 10 in its presence.

As in earlier years, the efficacy of good problems is marred by the preponderance of easy questions. At the risk of repetition, it is time that the number of questions in Paper 1 is reduced to about 15 or 20. Also questions seeking numerical answers (without showing reasoning) can be asked as is done nowadays in JEE.

The leeway the papersetters get in Paper 2 raises expectations about the quality of the problems in it. This year three of them, viz. Q.1 about showing a cycle of length 3 in a tripartite graph, Q.3 about representing rationals as finite sums of reciprocals of factorials and Q.7 about locating the roots of a cubic are nice.

As in the past, there are several questions in Paper 1 which properly belonged to Paper 2 because the MCQ format (especially where the candidates are foretold that only one of the four options is correct), often masks whether a candidate who answers it correctly has got the correct thought.

Problem 3 of Paper 2 may be called the best question of the entire test. By the same token, Problem 17 of Paper 1 is the worst.